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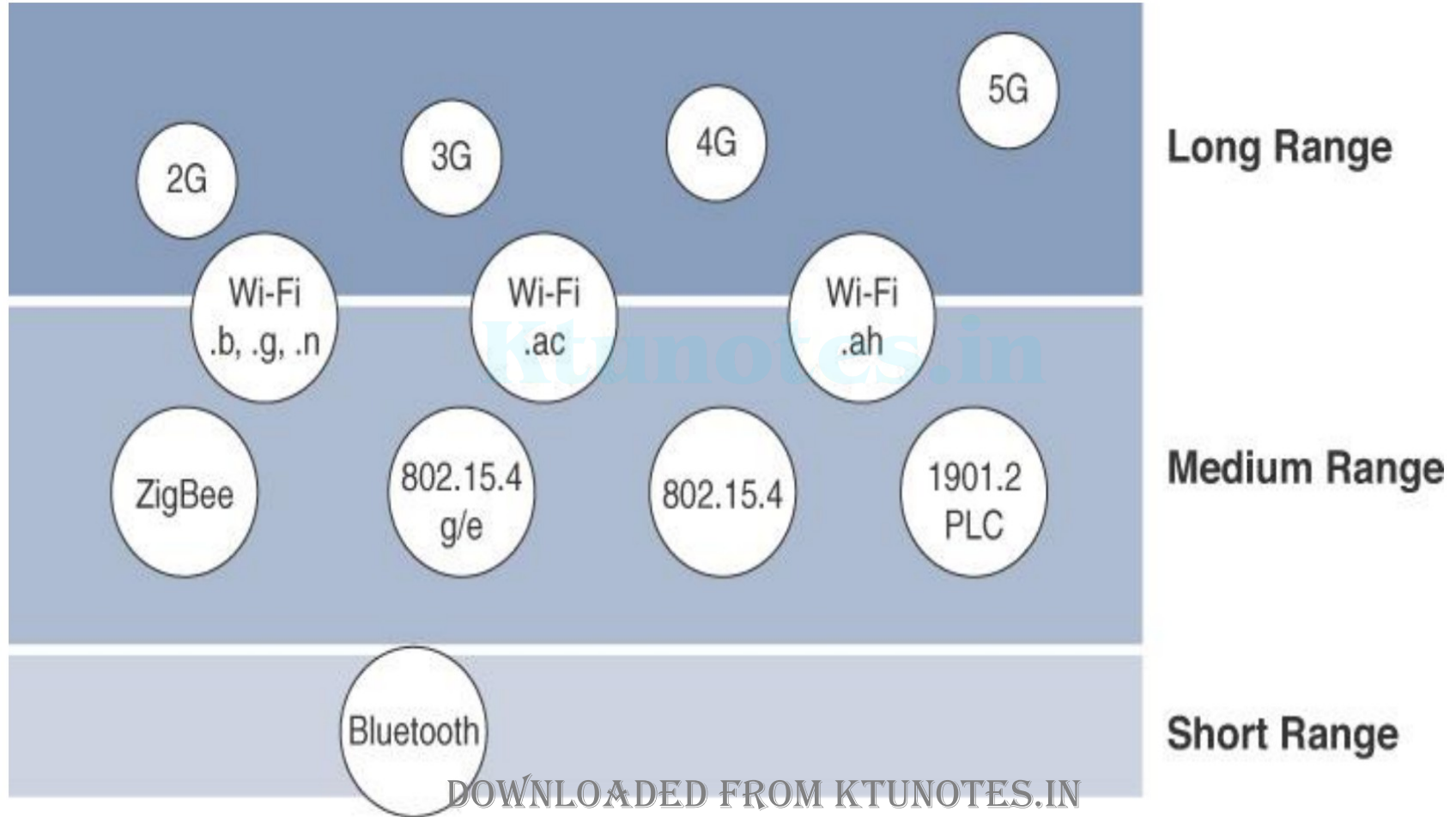
IoT Access Technologies

The methods for gaining access to an IoT system

- Wired/ wireless
- Wireless - Cellular Technologies or Non –Cellular Technologies
- Cellular - 2G, 3G, LTE etc
- Non –Cellular - WIFI, zigbee, LoRA etc

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The methods for gaining access to an IoT system



Common Information Set

- **Standardization and alliances:** The standards bodies that maintain the protocols for a technology
- **Physical layer:** The wired or wireless methods and relevant frequencies
- **MAC layer:** Considerations at the Media Access Control (MAC) layer, which bridges the **physical layer** with **data link control**
- **Topology:** The topologies supported by the technology
- **Security:** Security aspects of the technology

IEEE 802.15.4

- wireless access technology - wireless personal area networks
- low-cost and low-data-rate devices that are powered or run on batteries.
- Easy installation using a compact protocol stack while remaining both simple and flexible
- Applications - Home and building automation, Automotive networks, Industrial wireless sensor networks, Interactive toys and remote controls
- Eg.- ZigBee, 6LoWPAN etc

IEEE 802.15.4

Standardization and Alliances - IEEE 802.15 Task Group 4

→ They decide the PHY and MAC layer specifications

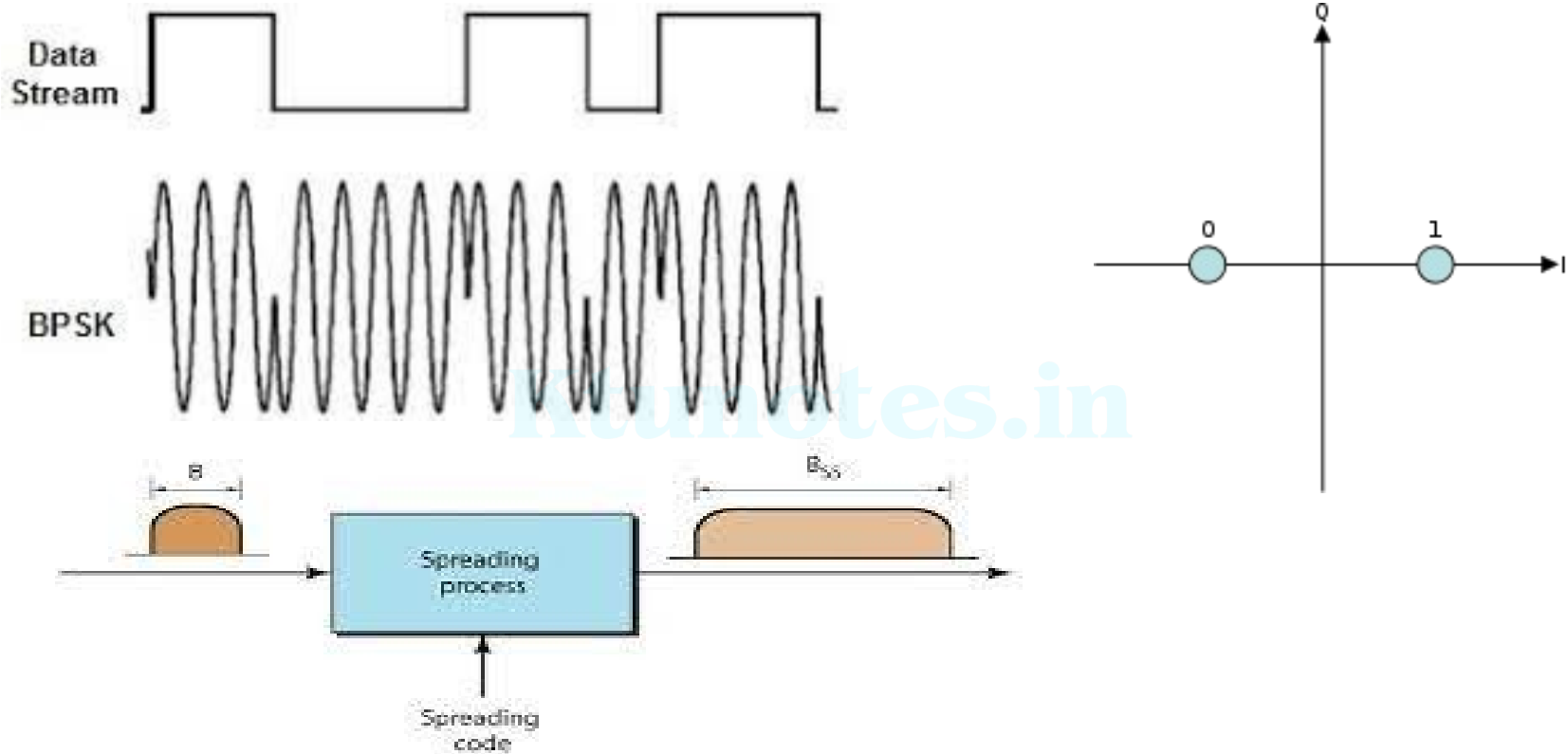
Physical Layer

→ 2.4 GHz, 16 channels, with a data rate of 250 kbps - world wide

→ **Modulation Techniques employed**

1) **BPSK PHY:** This is DSSS (Direct sequence spread spectrum) PHY, employing binary phase-shift keying (BPSK) modulation. BPSK specifies two unique phase shifts as its data encoding scheme.

IEEE 802.15.4



The spreading code is a series of numbers that look random, but are actually a pattern

IEEE 802.15.4

Physical Layer

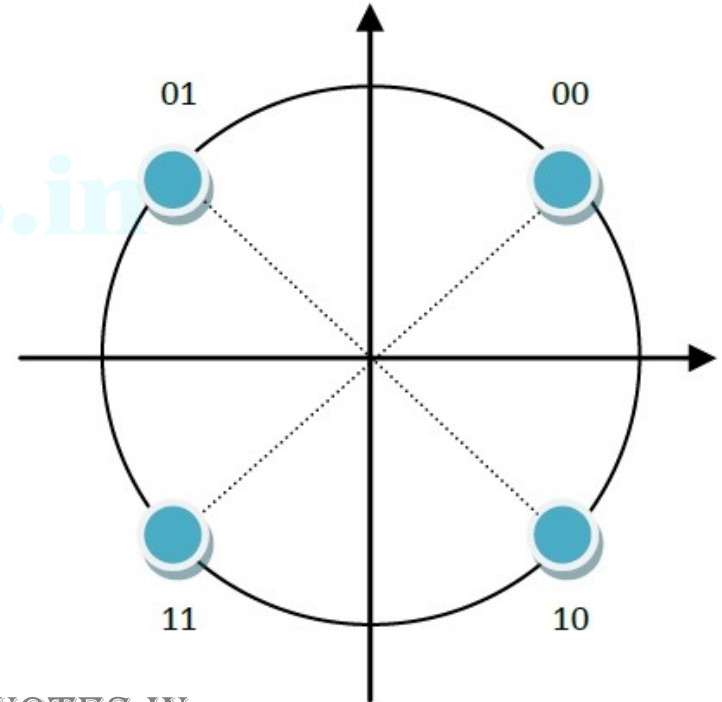
→ Modulation Techniques employed

2) OQPSK PHY - offset quadrature phase-shift keying modulation

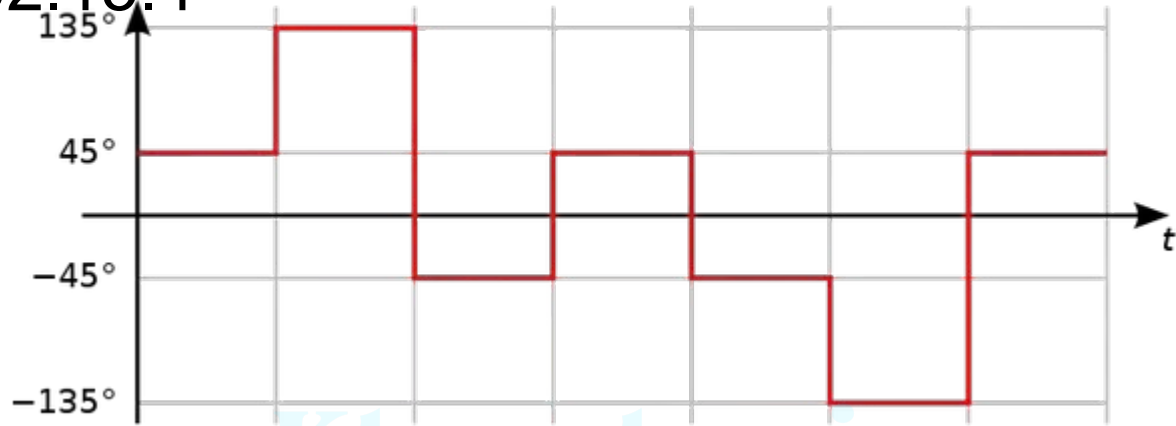
four unique bit values that are signaled by phase changes. An offset function that is present during phase shifts allows data to be transmitted more reliably.

IEEE 802.15.4

Quadrature Phase Shift Keying



IEEE 802.15.4 QPSK



IEEE 802.15.4

Physical Layer

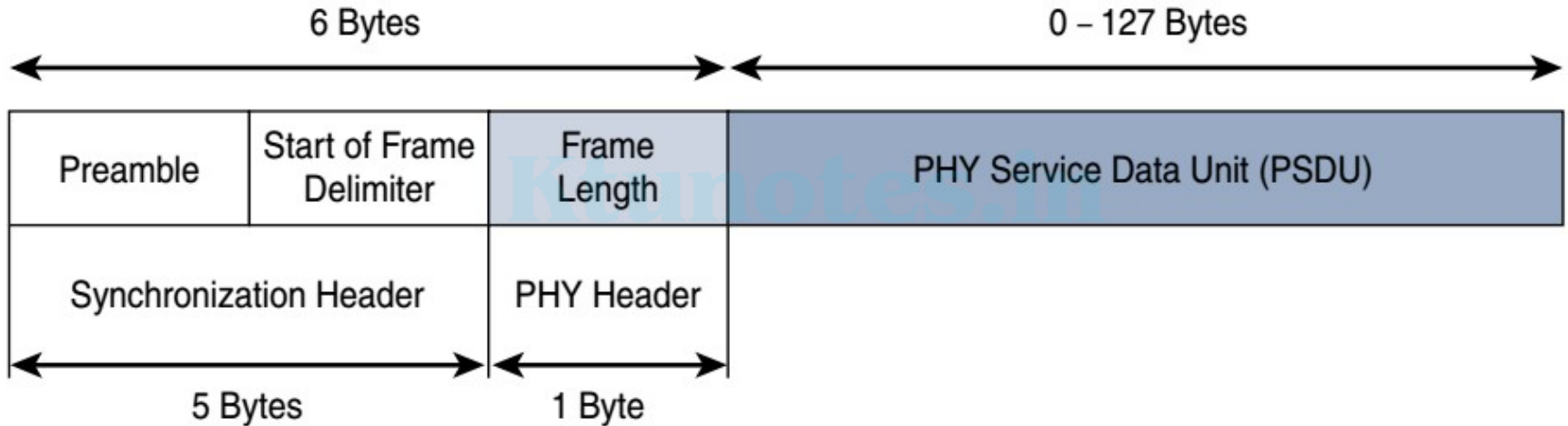
→ Modulation Techniques employed

2) **ASK PHY:** This is parallel sequence spread spectrum (PSSS) PHY. PSSS is an advanced encoding scheme that offers increased range, throughput, data rates, and signal integrity compared to DSSS. ASK uses amplitude shifts instead of phase shifts to signal different bit values.

IEEE 802.15.4

Physical Layer

→ IEEE 802.15.4 PHY Format



Preamble- sequence of bits for synchronization.

PHY Header - It lets the receiver know how much total data to expect

IEEE 802.15.4

MAC Layer

- **Manages** access to the PHY channel by defining how devices perform **multiple access**
- **Scheduling** and **routing** of data frames are coordinated
- **Network beaconing** (continuous transmission of small packets) for devices acting as coordinators
- PAN association and disassociation by a device
- Device security
- **Reliable** link communications between two peer MAC entities

IEEE 802.15.4

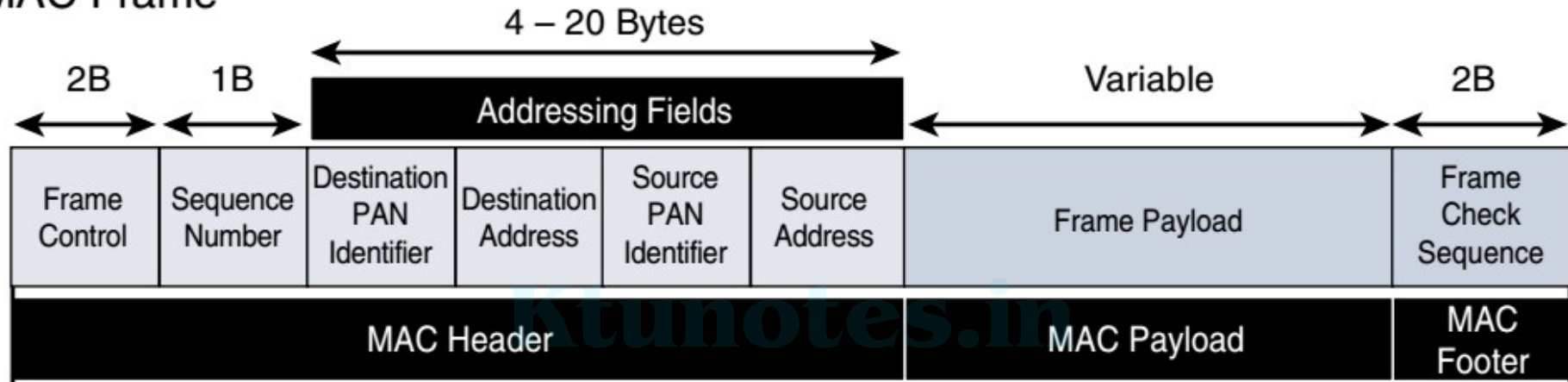
MAC Layer

Types of MAC frames

- **Data frame:** Handles all transfers of data
- **Beacon frame:** Used in the transmission of beacons from a PAN coordinator
- **Acknowledgement frame:** Confirms the successful reception of a frame
- **MAC command frame:** Responsible for control communication between devices

IEEE 802.15.4

MAC Frame



PHY Frame



IEEE 802.15.4

MAC Layer

- The MAC Header field is composed of the Frame Control, Sequence Number and the Addressing fields.
- The **Frame Control** field defines attributes such as **frame type**, **addressing modes**, and other **control flags**.
- The Sequence Number field indicates the sequence identifier for the frame.
- The Addressing field specifies the Source and Destination PAN Identifier fields as well as the Source and Destination Address fields.

IEEE 802.15.4

MAC Layer

- The MAC Payload field varies by individual frame type.
- Beacon frames have specific fields and payloads related to beacons,
- MAC command frames have different fields present.
- MAC Footer field is a frame check sequence (FCS) that is used by the receiving side to confirm the integrity of the data in the frame

IEEE 802.15.4

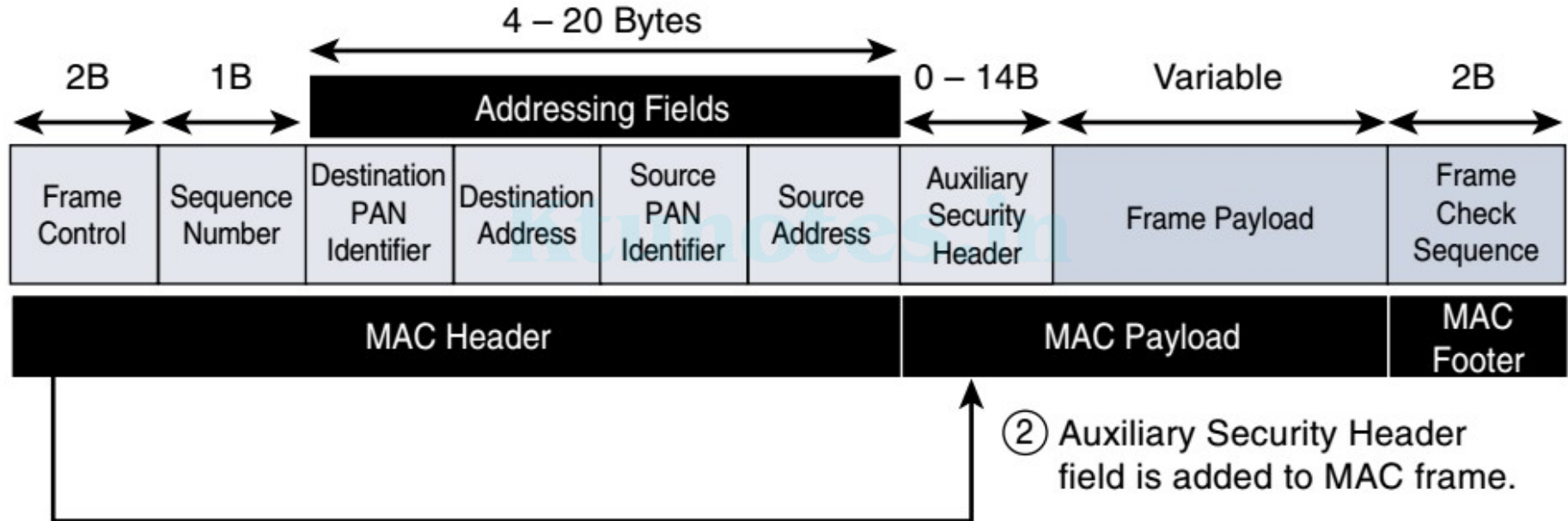
MAC Layer

Security

- The IEEE 802.15.4 specification uses **Advanced Encryption Standard** (AES) with a 128-bit key length as the base encryption algorithm for securing its data.
- AES in 802.15.4 also **validates** the data that is sent by a message integrity code (MIC), which is calculated for the entire frame using the same AES key that is used for encryption
- This is done by changing the frame format slightly and consumes some of the payload.

IEEE 802.15.4

MAC Layer - Security enabled frame format



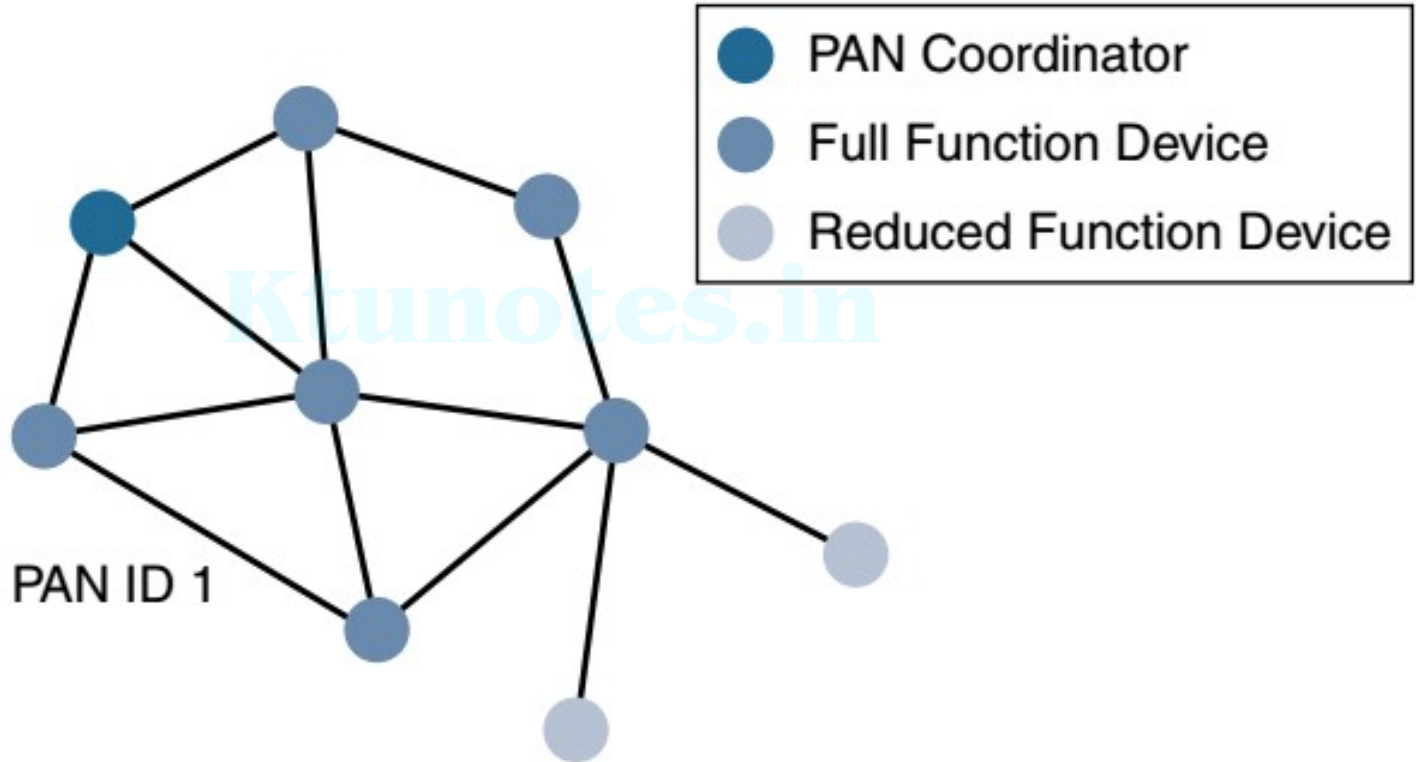
IEEE 802.15.4

Topology

- Networks can be built as star, peer-to-peer, or mesh topologies
- Reduced-function devices (RFDs) are confined to star topology
- A minimum of one Full-function devices(FFD) acting as a PAN coordinator is required to deliver services that allow other devices to associate and form a cell or PAN.
- Path selection within the MAC layer done at Layer 2 and is known as mesh-under
- All the nodes in the same 802.15.4 network should use the same PAN ID

IEEE 802.15.4

Topology



IEEE 802.15.4

- The IEEE frequently makes amendments to the core 802.15.4 specification, before integrating them into the next revision
- When it is done, a lowercase letter is appended.

IEEE 802.15.4e

- Expands the MAC layer feature for addressing problems related to **MAC reliability, unbounded latency, and multipath fading**
- To better cope with certain application domains, such as factory and process automation and smart grid

IEEE 802.15.4e

- MAC layer capabilities in the areas of frame format, security, determinism mechanism, and frequency hopping
- In IEEE 802.15.4e amendment is not applicable to the PHY layer, it is **pertinent to the MAC layer**
- This amendment enhances the MAC layer through various functions, which may be selectively enabled based on various implementations of the standard.

Enhancements to the MAC layer in IEEE 802.15.4e

- Time-Slotted Channel Hopping (TSCH) - utilizes different channels for transmission at different times

Each link will have different channel number in each cycle

Timeslot

Channel Offset

	0	1	2	3	4	5	6	7
0				A>C (ch13)			A>C	
1		A>B						
2					F>B			A>F B>F
3		C>D				A>D		
4			B>C					

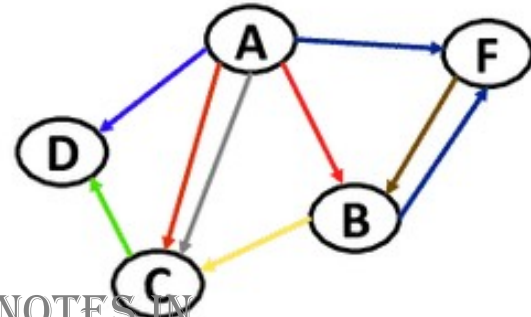
First Cycle k

	8	9	10	11	12	13	14	15
0				A>C (ch21)			A>C	
1		A>B						
2					F>B			A>F B>F
3		C>D				A>D		
4			B>C					

Second Cycle k+1

Allocated cell for the link between node C as a sender and node D as a receiver

Shared cells, in which more than one transmitter are scheduled to use a cell with a random back-off algorithm



IEEE 802.15.4e

- **TSCH** divides time into **fixed time periods**, or “time slots,” which offer guaranteed bandwidth and predictable latency. In a time slot, **one packet** and its **acknowledgement** can be transmitted, increasing network capacity because multiple nodes can communicate in the same time slot
- **Information elements (IEs)** allow for the exchange of information at the MAC layer in an extensible manner, either as header IEs (standardized) and/or payload IEs (private)

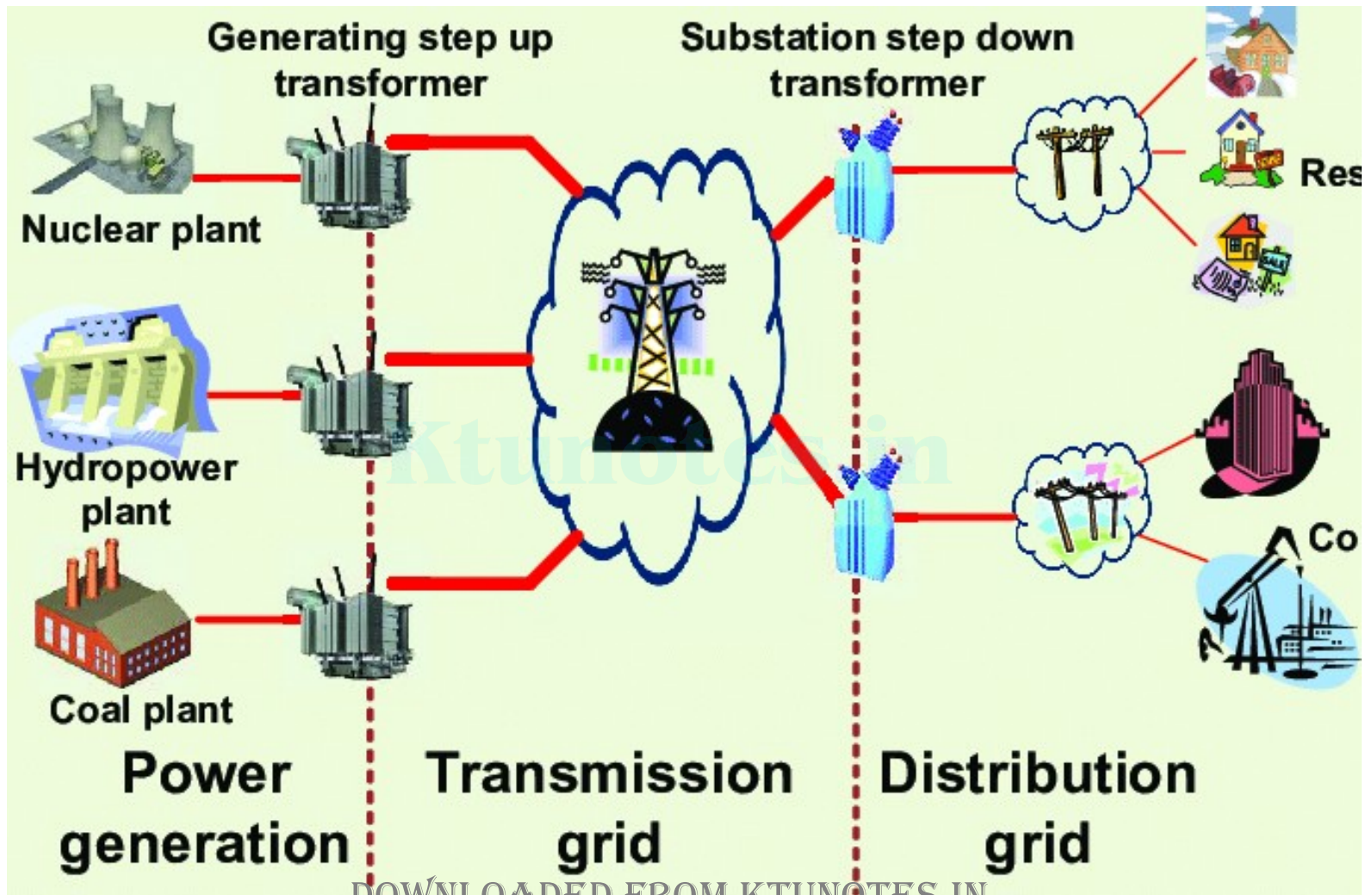
IEEE 802.15.4e

- **Enhanced beacons (EBs):** EBs extend the flexibility of IEEE 802.15.4 beacons to allow the construction of application-specific beacon content. This is accomplished by including relevant IEs in EB frames.
- **Enhanced beacon requests (EBRs):** The IEs in EBRs allow the sender to selectively specify the request of information.
- **Enhanced Acknowledgement:** It allows for the integration of a frame counter for the frame being acknowledged. It provides security when Acknowledgement frames are spoofed

IEEE 802.15.4g

IEEE 802.15.4g

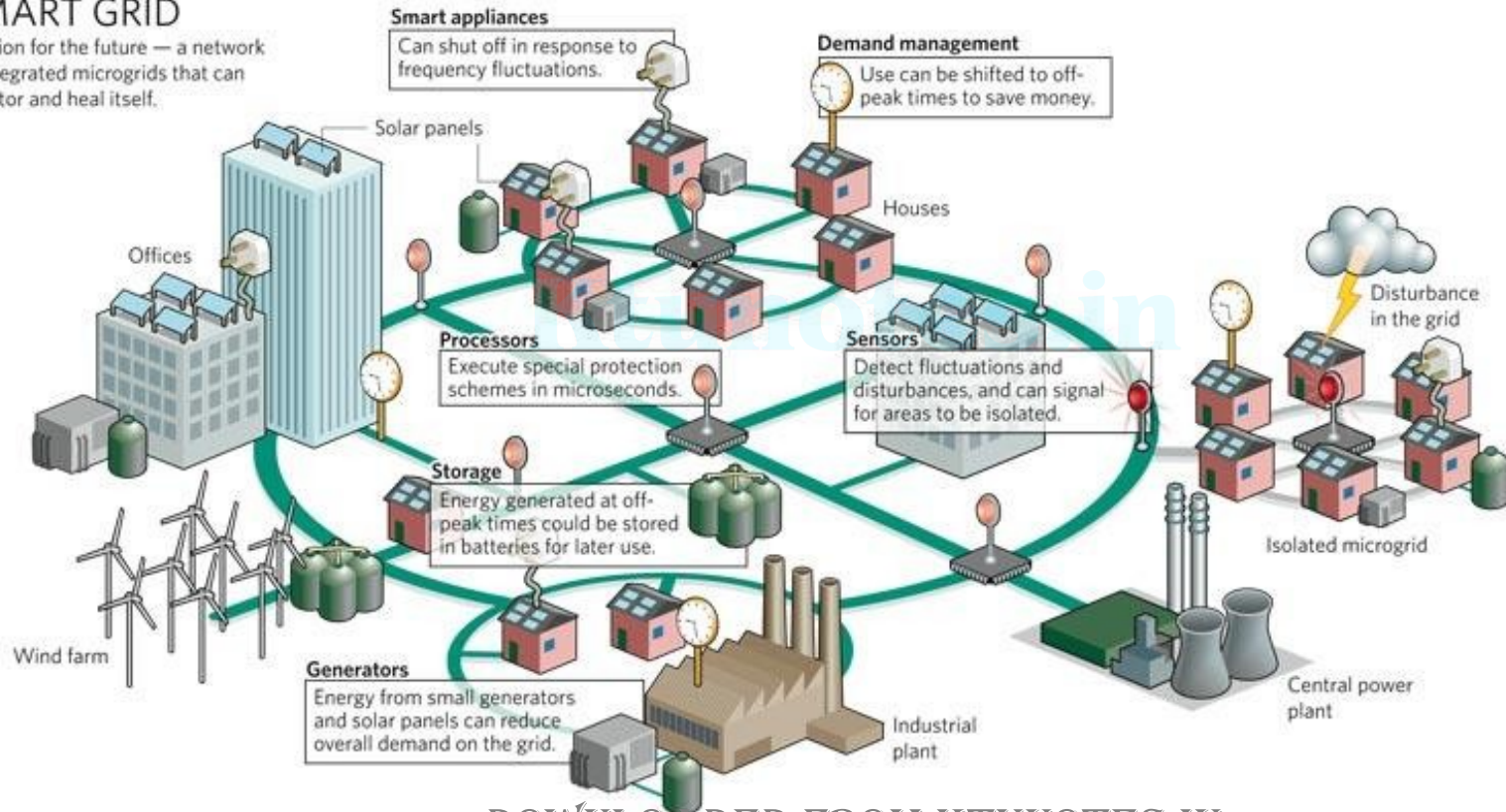
- The focus of this specification is the **smart grid** or, more specifically, **smart utility network communication**
- **New PHY definitions** are introduced, as well as some MAC **modifications** needed to **support** their implementation
- maximum PSDU or payload size of **127 bytes** was increased to **2047 bytes** -provides a better match for the **greater packet sizes** found in many **upper-layer** protocols.



Use real-time information to adjust electricity flows and ensure better energy efficiency

SMART GRID

A vision for the future — a network of integrated microgrids that can monitor and heal itself.



IEEE 802.15.4g

IEEE 802.15.4g

- **Supports multiple data rates** in bands ranging from 169 MHz to 2.4 GHz
- Modulation scheme followed were the following
 - **Multi-Rate and Multi-Regional Frequency Shift Keying (MR-FSK)**: Offers good transmit power efficiency due to the constant envelope of the transmit signal
 - **Multi-Rate and Multi-Regional Orthogonal Frequency Division Multiplexing (MR-OFDM)**: Provides higher data rates but may be too complex for low-cost and low-power devices

IEEE 802.15.4g

IEEE 802.15.4g

■ Multi-Rate and Multi-Regional Offset Quadrature Phase-Shift Keying (**MR-OQPSK**): making multi-mode systems more cost-effective and easier to design

→ In **IEEE 802.15.4g** majority of changes were made only in **PHY layer** and not **MAC**

IEEE 802.15.4g

IEEE 802.15.4g

■ Multi-Rate and Multi-Regional Offset Quadrature Phase-Shift Keying (MR-OQPSK): making multi-mode systems more cost-effective and easier to design

→ In **IEEE 802.15.4g** majority of changes were made only in **PHY layer** and not **MAC**

IEEE 802.15.4

IEEE 802.15.4g

Applications

- Distribution automation and industrial supervisory control and data acquisition (SCADA)

Public lighting, Environmental wireless sensors in smart cities
Electrical vehicle charging stations, Smart parking meters,
Microgrids, Renewable energy

IEEE 802.15.4

Security

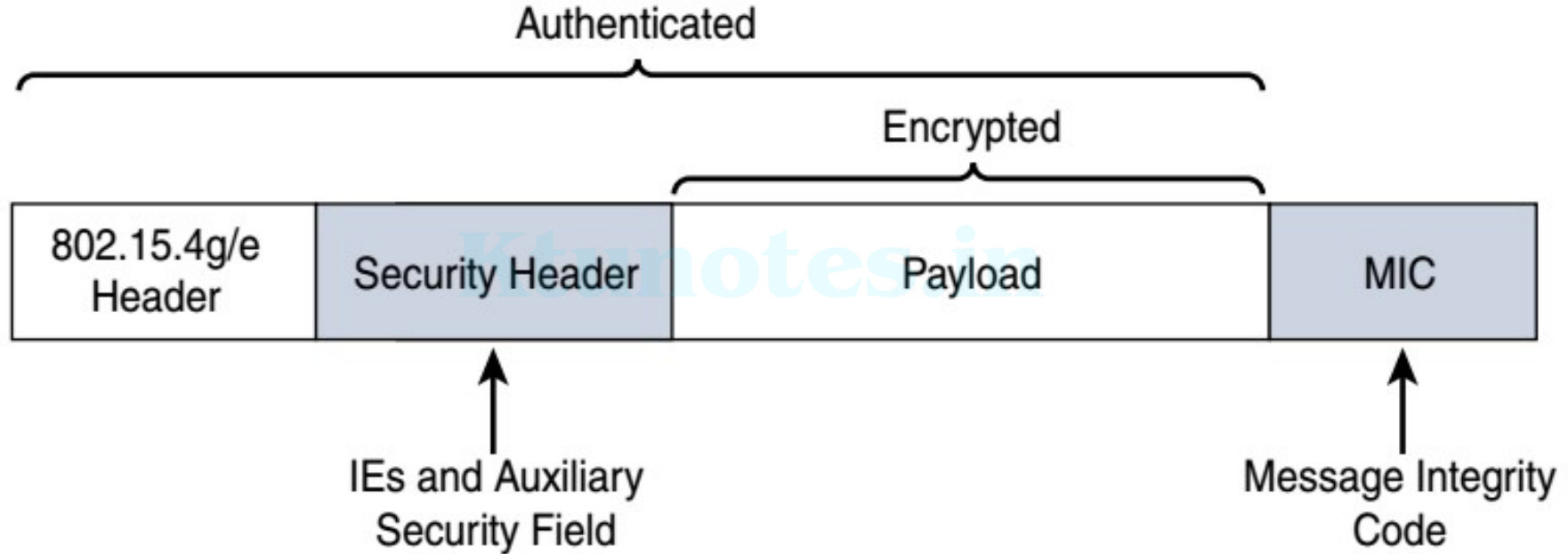
Both IEEE 802.15.4g and 802.15.4e **inherit** their security attributes from the **IEEE 802.15.4-2006** specification.

The encryption is provided by **AES**, with a **128-bit key**

In addition to the Auxiliary Security Header field initially defined in 802.15.4-2006, a **secure acknowledgement** and a **secure Enhanced Beacon field** complete the MAC layer security

IEEE 802.15.4

Security

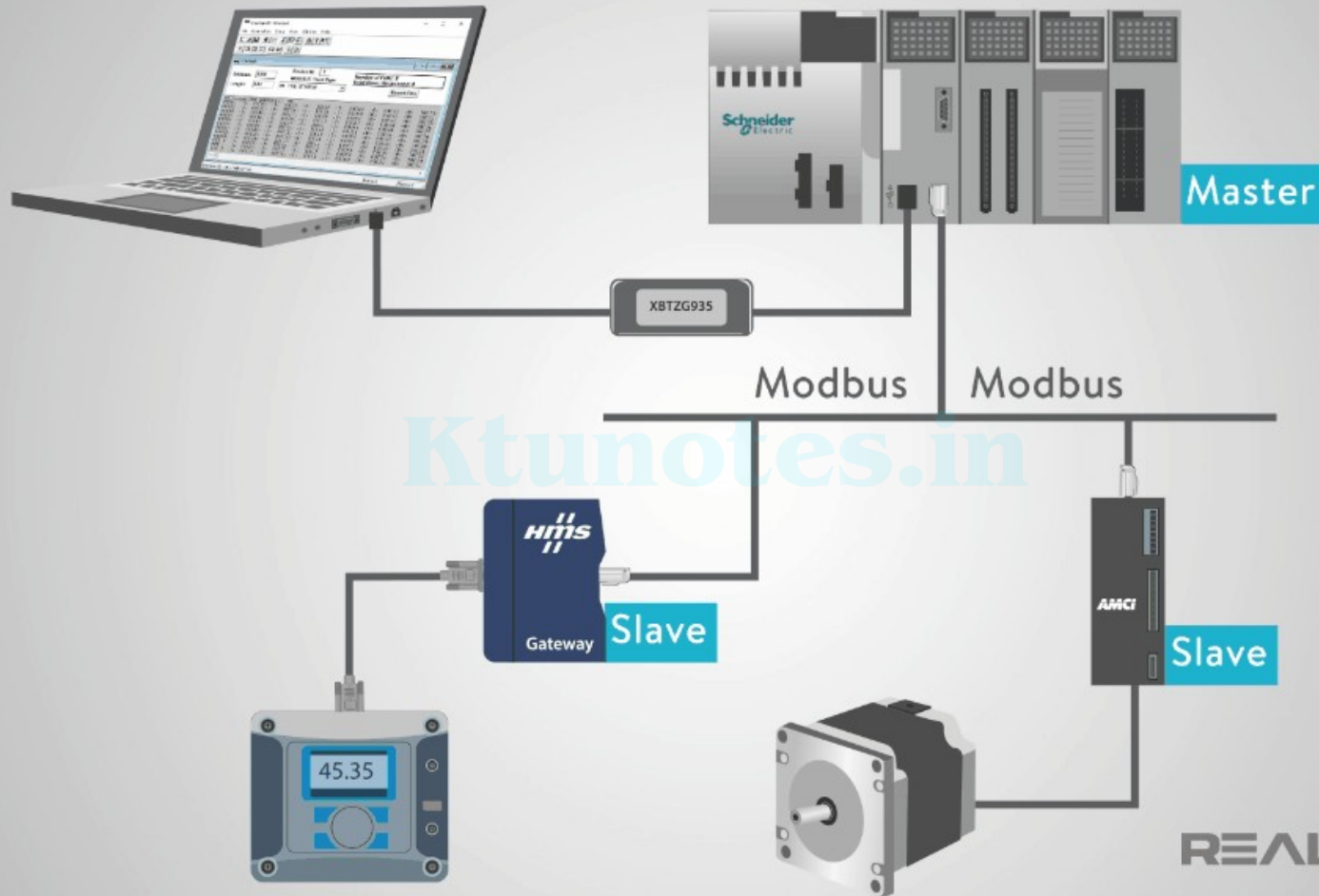


Modbus

- It is an **industrial protocol** that was developed in 1979 to make **communication** possible between **automation devices**
- It was created by the first programmable logic controller (PLC) vendor, **Modicon**
- It is a **request-response** protocol implemented using a **master-slave** relationship
- In a master-slave relationship, communication always occurs in pairs—one **device must initiate a request and then wait for a response**—and the initiating device (the master) is responsible for initiating every interaction

Modbus

- Typically, the **master** is a human machine interface (**HMI**) or Supervisory Control and Data Acquisition (**SCADA**) system and the **slave** is a **sensor**, programmable logic controller (**PLC**), or programmable automation controller (PAC)
- Modbus is transmitted over serial lines between devices. The simplest setup would be a single serial cable connecting the serial ports on two devices, a Client and a Server.



ZigBee

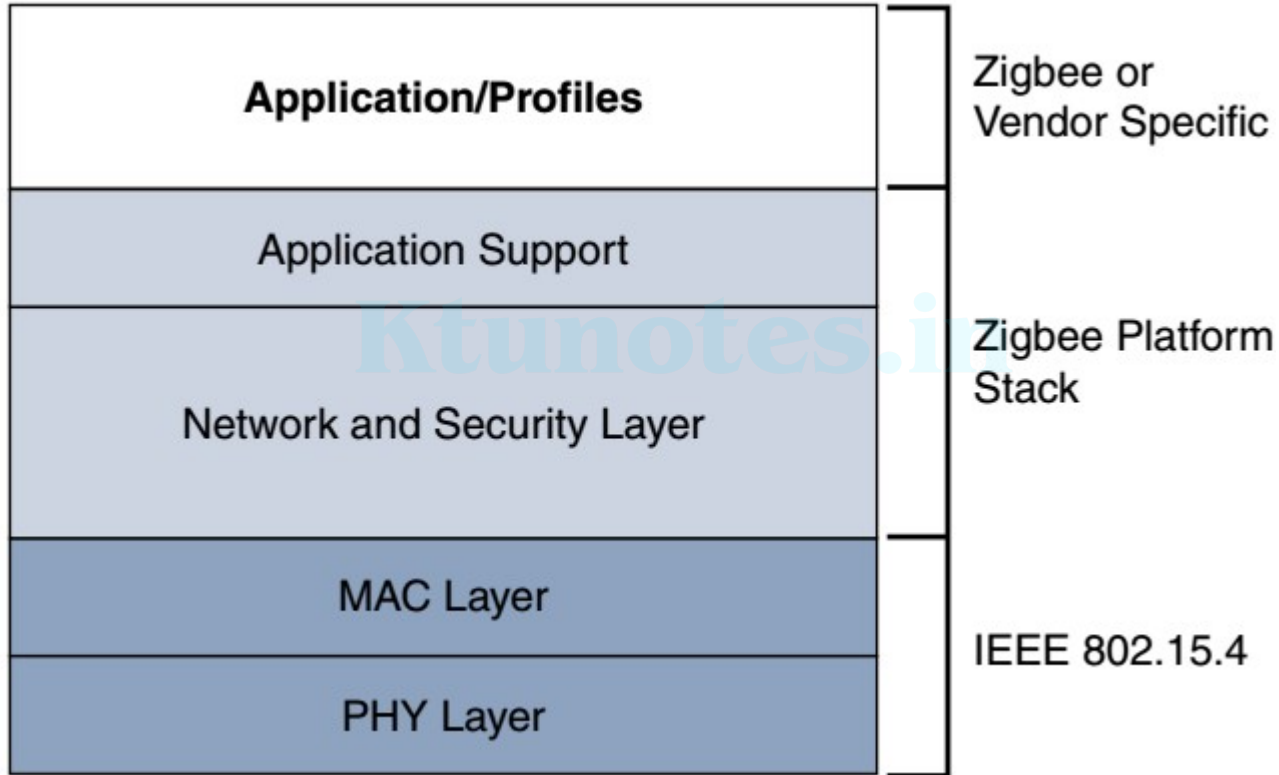
- It is a Protocol Stack Utilizing **IEEE 802.15.4**
- Promoted through the **ZigBee Alliance**, ZigBee defines **upper-layer components** (network to application) as well as application profiles.
- Common profiles include building automation, home automation, and healthcare
- ZigBee also **defines device object functions**, such as device role, device discovery, network join, and security
- ZigBee solutions are aimed at smart objects and sensors that have **low bandwidth** and **low power** needs

ZigBee

Applications

- **Industrial and commercial automation** - measuring temperature and humidity to tracking assets
- **Home automation** - control lighting, thermostats, and security functions
- **ZigBee Smart Energy** - brings together a variety of interoperable products, such as smart meters, that can monitor and control the use and delivery of utilities, such as electricity and water

ZigBee Protocol Stack (Architecture)



ZigBee Protocol Stack

- ZigBee utilizes the **IEEE 802.15.4** standard at the lower **PHY** and **MAC** layers
- ZigBee **specifies** the **network** and **security** layer and **application** support layer that sit on top of the lower layers
- The network and security layer provides mechanisms for **network startup, configuration, routing, and securing** communications.
- This includes **calculating routing paths** in what is often a changing topology, **discovering neighbors**, and **managing the routing tables** as devices join for the first time.

ZigBee Protocol Stack

- The **network layer** is also responsible for forming the appropriate **topology**, which is often a mesh but could be a star or tree as well.
- It uses the Advanced Encryption Standard (**AES**) with a **128-bit** key and also provides security at the network and application layers
- ZigBee predefines many **application profiles** for certain industries, and vendors can optionally create their **own custom** ones at this layer.
- Home Automation and Smart Energy are two examples of popular application profiles

ZigBee Protocol Stack

- ZigBee uses Ad hoc **On-Demand** Distance Vector (**AODV**) routing across a mesh network.
- This routing algorithm **does not send a message until a route is needed**.
- Assuming that the next hop for a route is not in its routing table, a network node broadcasts a request for a routing connection. This causes a burst of routing-related traffic, but after a comparison of various responses, the path with the lowest number of hops is determined for the connection

LoRaWAN

- It is a LPWAN (Low-Power Wide-Area) protocol designed to connect **battery operated** 'things' to the internet particularly well adapted for **long-range**
- **LoRa (Long Range Radio)** was developed by a French company named **Cycleo**.
- Later was acquired by **Semtech**
- It is a wireless modulation technique for long range communication



LoRaWAN

→ LoRaWAN Layers

LoRa Alliance	Applications				
	CoAP	MQTT	IPv6/ 6LoWPAN	Raw	Others
	LoRaWAN MAC				
	LoRa PHY Modulation				
	868MHz	915MHz	Other Regional Bands		

LoRaWAN

→ Physical Layer

Semtech LoRa **modulation** is based on **chirp spread spectrum modulation** which trades a **lower data rate** for receiver sensitivity to significantly **increase** the communication **distance**

it allows demodulation below the noise floor, **offers robustness to noise and interference**, and manages a **single channel** occupation by **different spreading factors**

LoRaWAN

→ Physical Layer

Here, **each bit is spread by chipping factor.**

The number of chips per bit is known as spreading factor.

In other words, the **duration of chirp** is known as **spreading factor**.

This enables LoRa devices to receive on **multiple channels in parallel**

LoRaWAN

→ Physical Layer

LoRaWAN 1.0.2 regional specifications describe the use of the main unlicensed sub-GHz frequency bands of 433 MHz, 779–787 MHz, 863–870 MHz, and 902–928 MHz, as well as regional profiles for a subset of the 902–928 MHz bandwidth.

For example, Australia utilizes 915–928 MHz frequency bands, while South Korea uses 920–923 MHz and Japan uses 920–928 MHz.

LoRaWAN

→ LoRa gateways

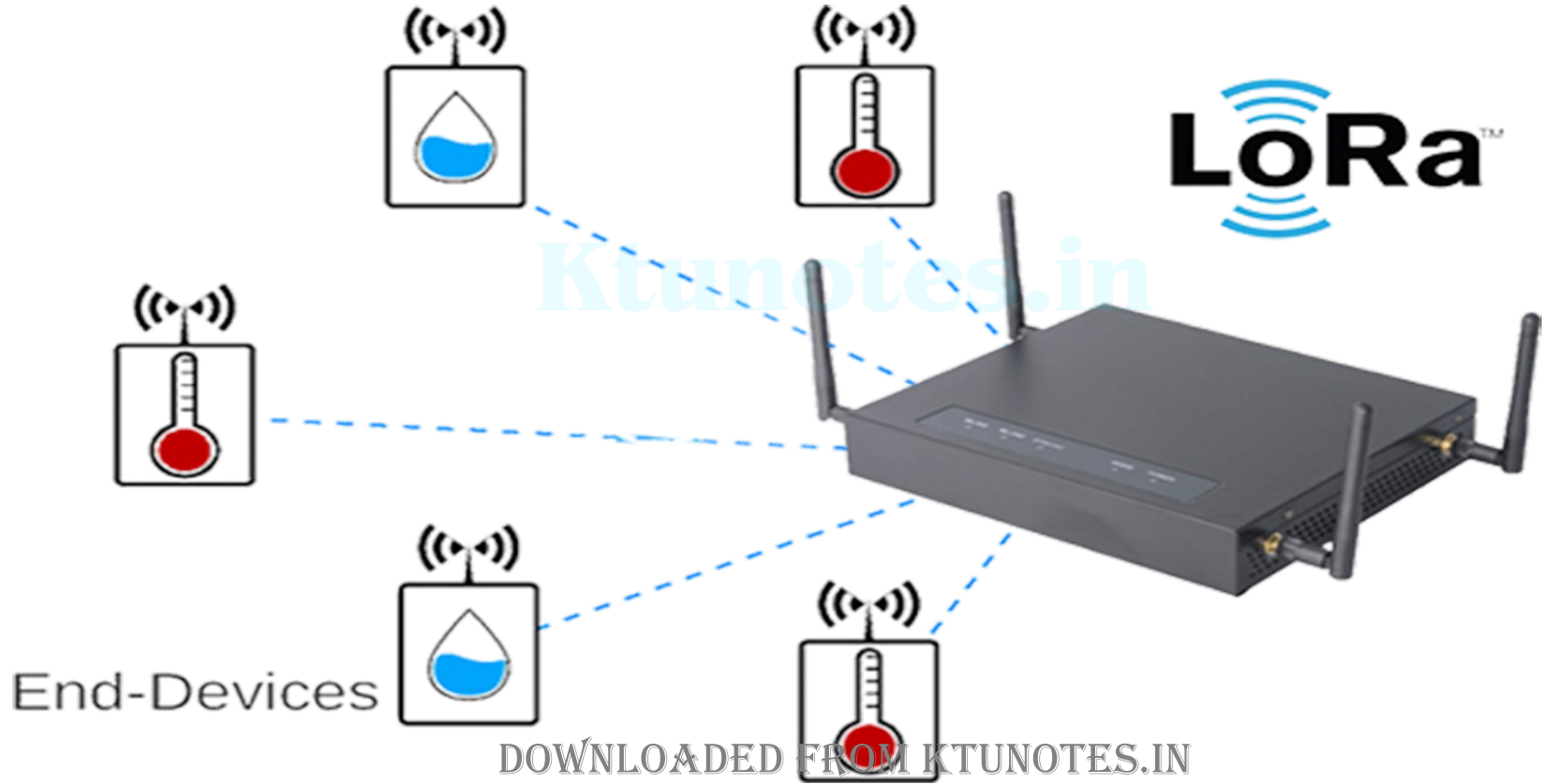
It is **deployed** as the **center hub** of a **star network** architecture

It uses multiple transceivers and channels and can demodulate multiple channels at once or even demodulate multiple signals on the same channel simultaneously.

LoRa gateways serve as a **transparent bridge relaying data** between endpoints, and the endpoints use a single-hop wireless connection to communicate with one or many gateways.

LoRaWAN

→ LoRa gateways



LoRaWAN

→ LoRa gateways

The data rate in LoRaWAN varies depending on the frequency bands and adaptive data rate (**ADR**).

ADR is an **algorithm** that manages the **data rate** and **radio signal** for each **endpoint**.

It **ensures** that packets are delivered at the **best data rate possible** and that network performance is both optimal and scalable.

LoRaWAN

→ LoRa gateways - ADR algorithm

Endpoints **close** to the **gateways** with **good signal values** transmit with the **highest data rate**, which enables a **shorter transmission time** over the wireless network, and the **lowest transmit power**. Meanwhile,

endpoints at the **edge** of the link budget communicate at the **lowest data rate** and highest transmit power

Data rate - distance - signal strength- power

LoRaWAN

An important feature of LoRa is its ability to handle **various data rates** via the **spreading factor**.

The spreading factor defines the relation between **symbol rate** and **chip rate**.

Chip rate is the **number of bits per second** (chips per second) **used in the spreading signal**.

LoRaWAN

A **higher spreading factor increases** sensitivity and **range**, but also **prolongs the airtime** of a packet and will likely raise the risk of a **collision**

Devices with a **low** spreading factor (SF) achieve **less distance** in their communications but transmit at **faster speeds**, resulting in **less airtime**.

A **higher SF** provides **slower transmission** rates but achieves a **higher reliability** at **longer** distances.

LoRaWAN

For example, at an SF value of 12 for 125 kHz of channel bandwidth, the data rate is 250 bps. However,

when the SF is decreased to a value of 7, the data rate increases to 5470 bps

The effect of increasing the bandwidth is that faster data rates can be achieved for the same spreading factor.

LoRaWAN

MAC Layer

- This layer takes advantage of the LoRa physical layer and classifies LoRaWAN endpoints to optimize their battery life and ensure downstream communications to the LoRaWAN endpoints.
- The LoRaWAN specification documents three classes of LoRaWAN devices

LoRaWAN

MAC Layer

Class A: This class is the **default implementation**. Optimized for **battery-powered nodes**.

It allows **bidirectional** communications, where a given node is able to **receive** downstream traffic **after transmitting**.

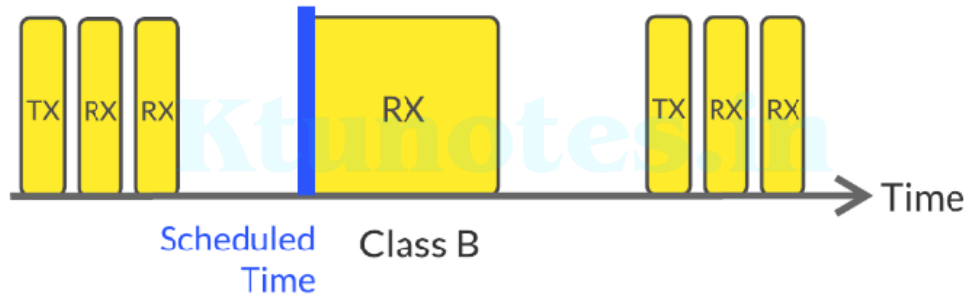
Two receive windows are available after each transmission.

LoRaWAN

MAC Layer

Class B: This class was designated “experimental” in LoRaWAN 1.0.1 until it can be better defined. A Class B node or endpoint should get additional receive windows compared to Class A, but gateways must be synchronized through a beaconing process

Class C: This class is particularly adapted for **powered nodes**. This classification enables a node to be **continuously listening** by keeping its **receive window open when not transmitting**

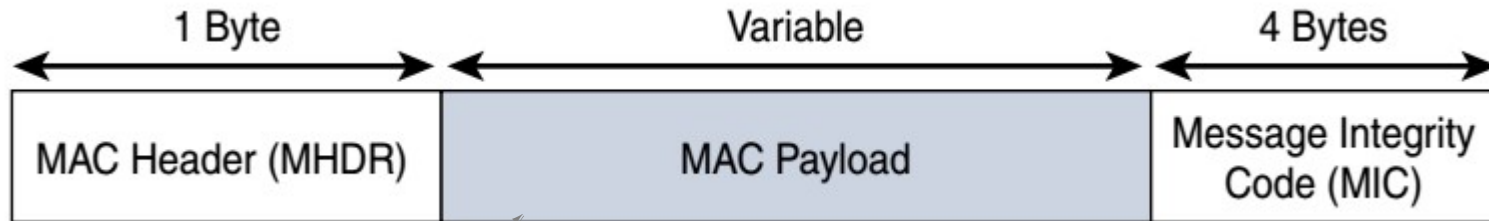


LoRaWAN

MAC Layer

LoRaWAN messages, either uplink or downlink, have a PHY payload composed of a 1-byte MAC header, a variable-byte MAC payload, and a MIC that is 4 bytes in length.

The MAC **payload size** depends on the **frequency band** and the data rate, ranging from 59 to 230 bytes for the 863–870 MHz band and 19 to 250 bytes for the 902–928 MHz band



LoRaWAN

LoRaWAN utilizes six MAC message types

1. join request and join accept messages for over-the-air (OTA) activation and joining the network
2. Unconfirmed data - need not be acknowledged
3. Confirmed data - messages need to be acknowledged
4. Uplink data- messages are sent from endpoints to the network server
5. Downlink - messages flow from the network server to a single endpoint
6. Multicast

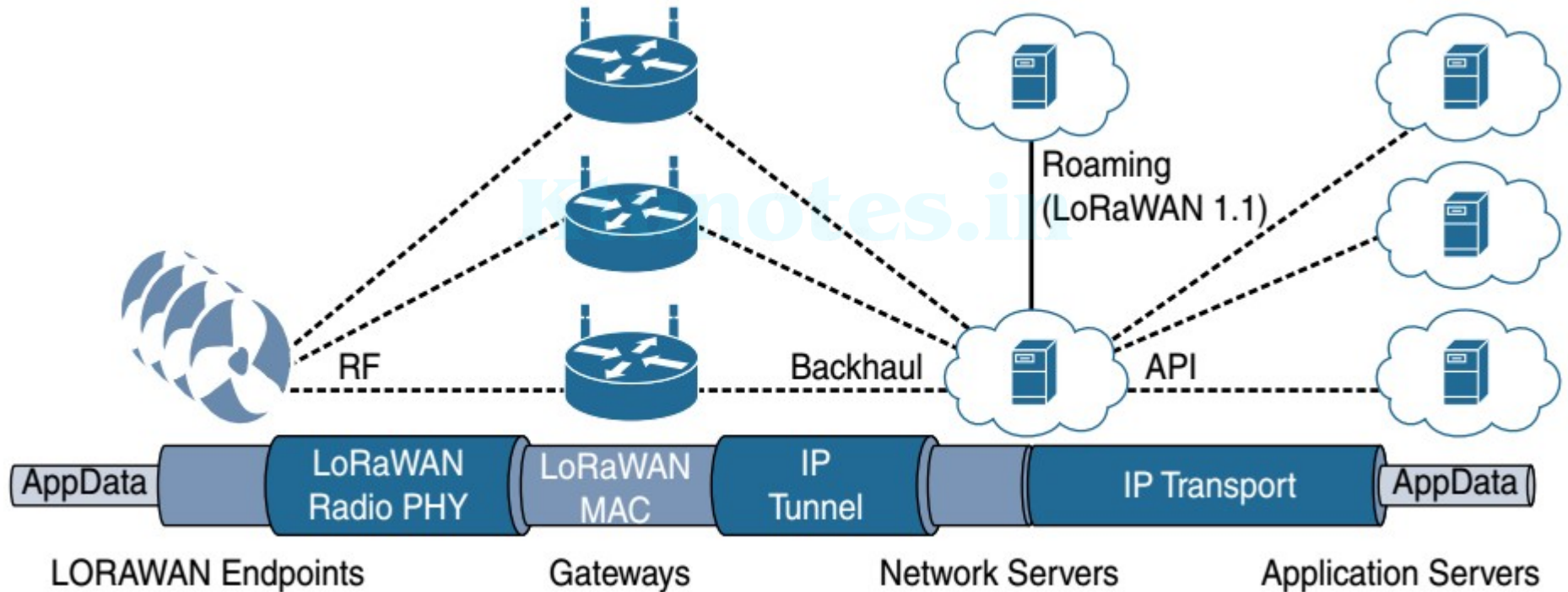
LoRaWAN

LoRaWAN endpoint addressing

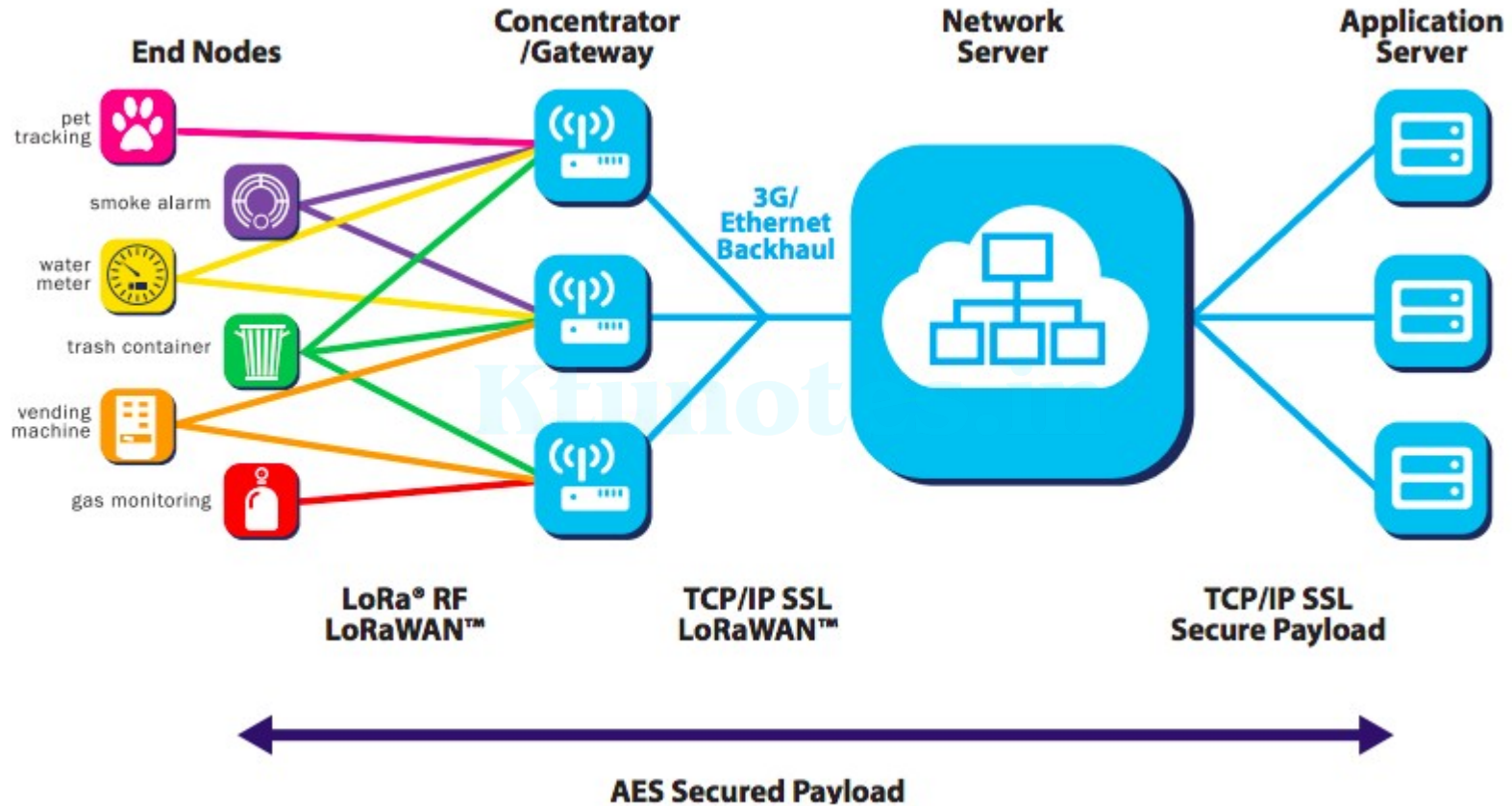
- A global end device ID or DevEUI represented as an IEEE EUI-64 address
- A global application ID or AppEUI represented as an IEEE EUI-64 address that uniquely identifies the application provider
- By their end device address, known as a DevAddr, a 32-bit address . The **7 most significant bits** are the **network identifier** (NwkID), which identifies the LoRaWAN network. The **25 least significant bits** are used as the network address (NwkAddr) to identify the **endpoint** in the network

LoRaWAN

Topology - star of stars



LoRaWAN



LoRaWAN

Topology - star of stars

The infrastructure consists of **endpoints exchanging packets** through gate ways acting as bridges, with a **central LoRaWAN network server**.

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Gateways connect to the backend network using standard IP connections, and endpoints communicate directly with one or more gateways

LoRaWAN endpoints transport their application data over the LoRaWAN MAC layer on top of one of the supported PHY layer frequency bands

LoRaWAN

Topology - star of stars

- The **application data** is contained in upper protocol layers which is **not** managed by the LoRa Alliance
- These upper layers could just be raw data on top of the LoRaWAN MAC layer, or the data could be stacked in multiple protocols like MQTT, CoAP etc.
- LoRaWAN gateways act as bridges that relay between endpoints and the network servers.
- Multiple gateways can receive and transport the same packets

LoRaWAN

Security

- LoRaWAN endpoints must implement **two layers of security**
- The first is “**network security**” applied at the **MAC layer**, guarantees the **authentication** of the **endpoints**.
- Also, it **protects LoRaWAN packets** by performing encryption based on **AES**

LoRaWAN

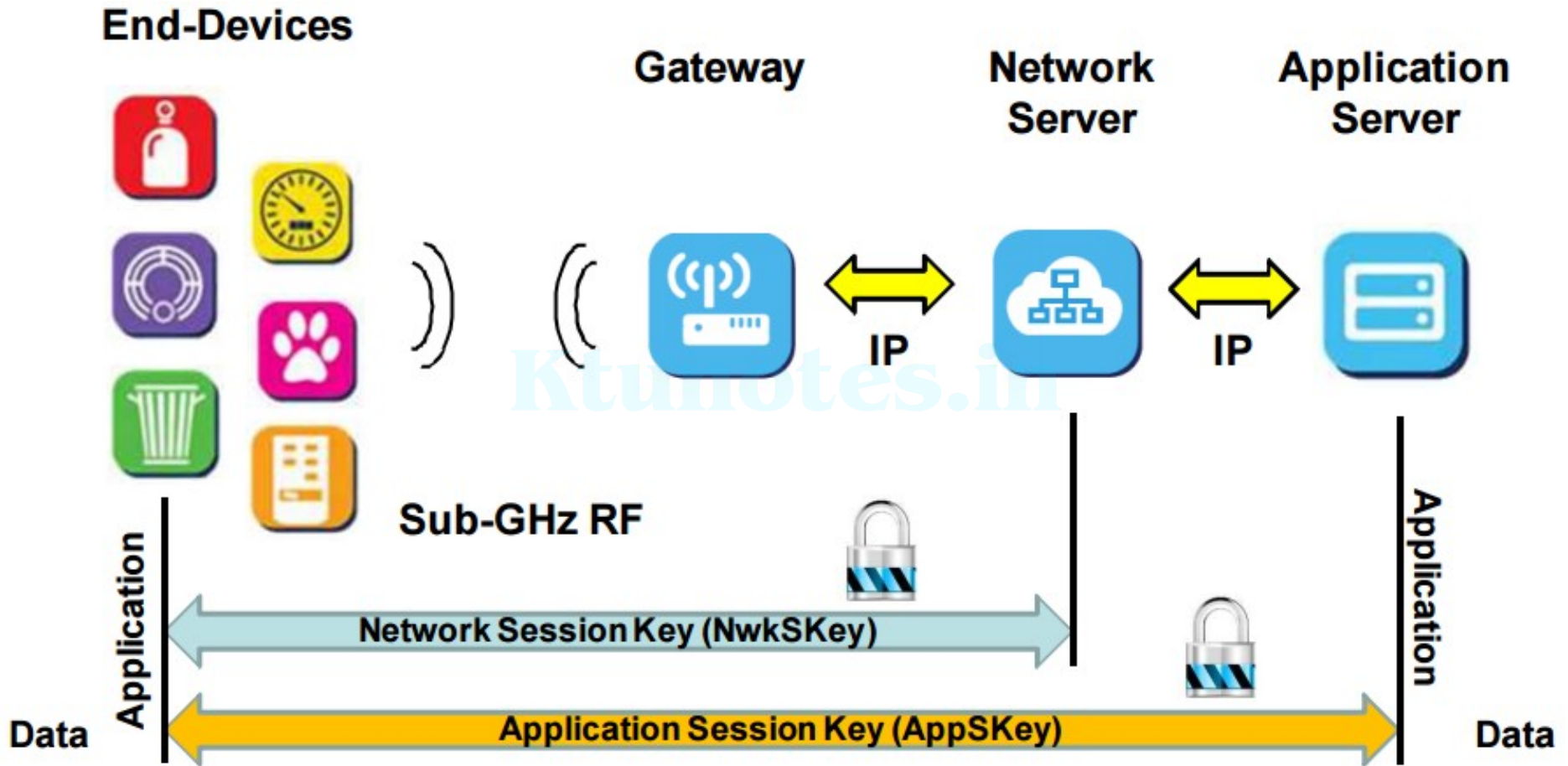
Security

- Each endpoint implements a **network session key** (NwkSKey), used by both itself and the LoRaWAN network server.
- It ensures **data integrity** through computing and checking the MIC (Message Integrity Check) of every data message as well as encrypting and decrypting MAC-only data message payloads.

LoRaWAN

Security

- The **second layer** is an **application session key** (AppSKey), which performs encryption and decryption functions **between the endpoint and its application server**. Furthermore, it computes and checks the **application-level MIC**, if included. This ensures that the LoRaWAN service provider does not have access to the application payload if it is not allowed that access
- Endpoints receive their AES-128 application key (AppKey) from the application owner



IoT and Cellular networks

- Existing cellular technologies, such as GPRS, Edge, 3G, and 4G/LTE, are **not** particularly well **adapted** to **battery-powered** devices and **small objects** specifically developed for the IoT
- IoT requires **low throughput, low power consumption, decrease the complexity and cost of devices**
- This resulted in the definition of the **LTE-M** work item
- But LTE-M device category was not sufficiently close to LPWA capabilities, Narrowband IoT (**NB-IoT**) was proposed

IoT and Cellular networks

Standardization and Alliances

- 3rd Generation Partnership Project (3GPP) is the standardization organization cellular networks
- Many service providers and vendors make up 3GPP
- IoT-related contribution is handled by either 3GPP or the GSM EDGE Radio Access Networks (GERAN) group

IoT and Cellular networks

LTE-MTC or LTE-M -Long Term Evolution Machine Type Communication

- The first enhancements to better support IoT devices in 3GPP occurred in LTE Release 12 - A new user equipment (UE) category, Category 0, was added
- It includes important characteristics to be supported by both the network and end devices
- These Cat 0 characteristics include Power saving mode (PSM) and Half-duplex mode

LTE-M

Characteristics of the LTE-M

- **Lower receiver bandwidth:** Bandwidth has been lowered to 1.4 MHz vs usual 20 MHz. This further simplifies the LTE endpoint.
- **Lower data rate:** Data is around 200 kbps for LTE-M, compared to 1 Mbps for Cat 0
- **Half-duplex mode:** Just as with Cat 0, LTE-M offers a half-duplex mode that decreases node complexity and cost

LTE-M

Characteristics of the LTE-M

- **Enhanced discontinuous reception (eDRX)**: This capability increases from **seconds to minutes** the amount of time an endpoint can “**sleep**” **between paging cycles**.
- A paging cycle is a periodic check-in with the network.
- This extended “sleep” time between paging cycles **extends the battery lifetime** for an endpoint significantly

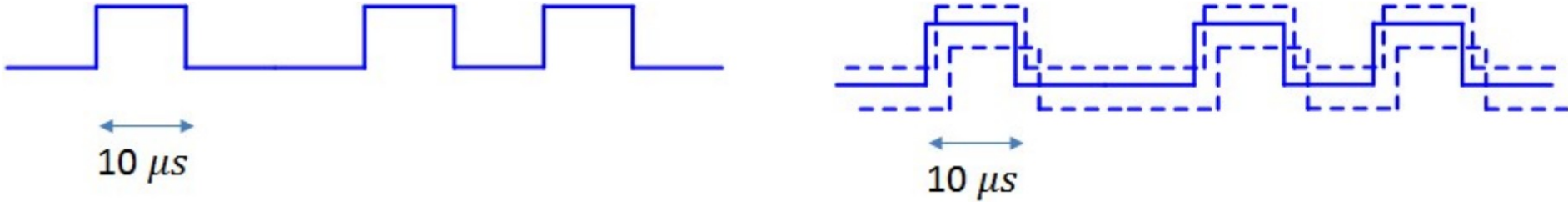
NB-IoT

- Recognizing that the definition of new LTE device categories was not sufficient to support LPWA IoT requirement, 3GPP specified Narrowband IoT (NB-IoT)
- Proposals pushed by various vendors before NB-IoT
 1. Extended Coverage GSM (EC-GSM), Ericsson
 2. Narrowband GSM (N-GSM), Nokia
 3. Narrowband M2M (NB-M2M), Huawei
 4. Narrowband OFDMA - Qualcomm
 5. Cooperative Ultra Narrowband (C-UNB), Sigfox
 6. Narrowband LTE (NB-LTE) - Alcatel-Lucent

NB-IoT

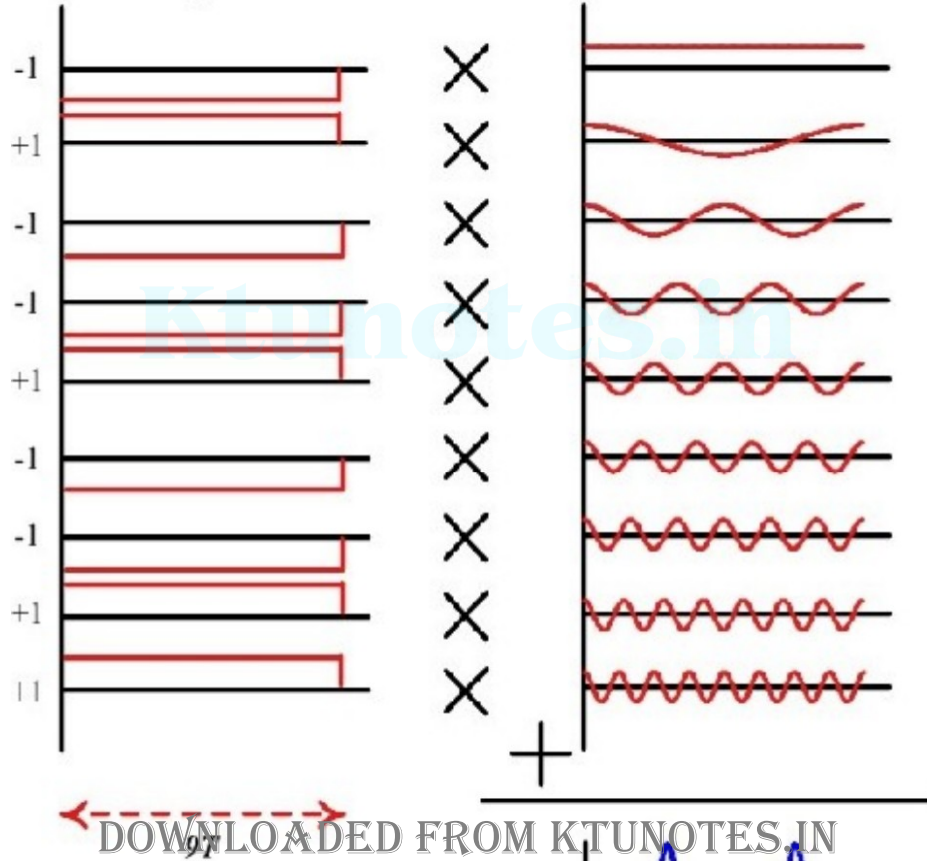
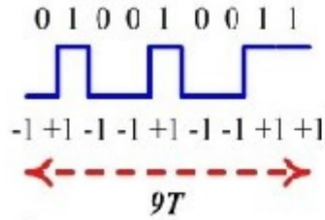
- Consolidation occurred with the agreement to specify a **single NB-IoT version** based on **orthogonal frequency-division multiple access (OFDMA)** in the downlink and a couple options for the uplink
- OFDMA is a modulation scheme in which individual users are assigned subsets of subcarrier frequencies. This enables multiple users to transmit low speed data simultaneously.

- Low data rate and multipath reception

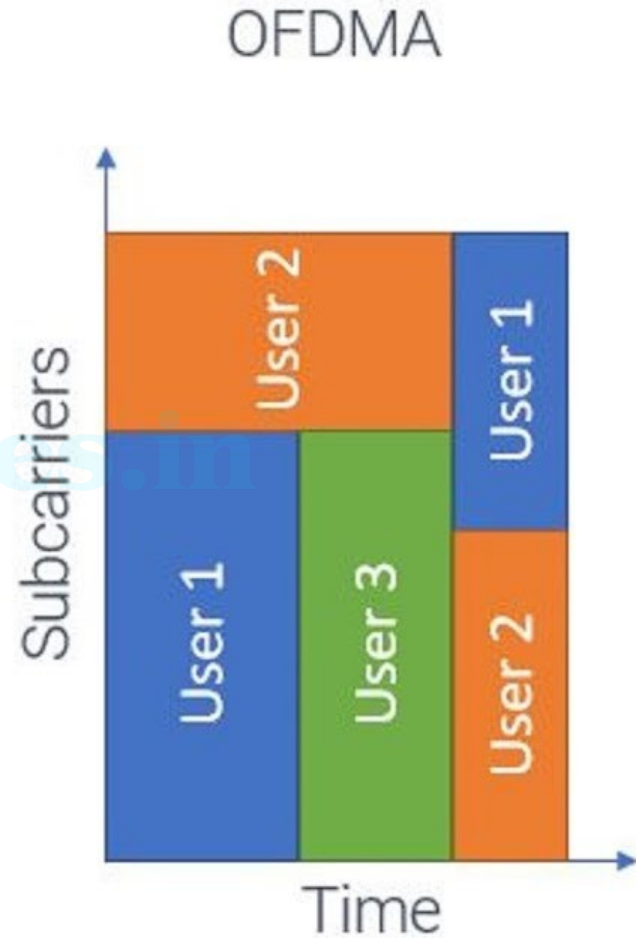
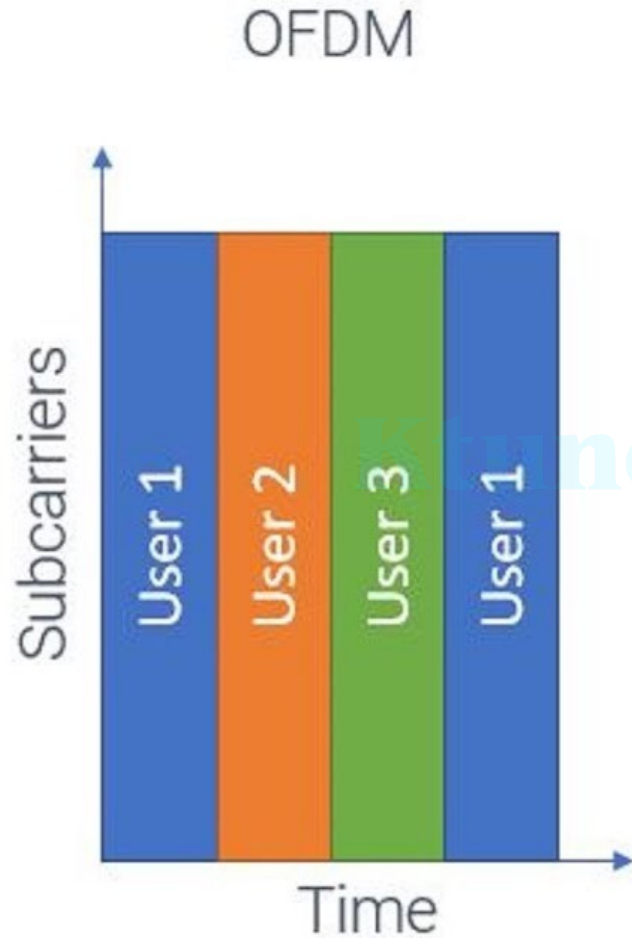


- High data rate and multipath reception





O F D M A



IEEE 802.15.4

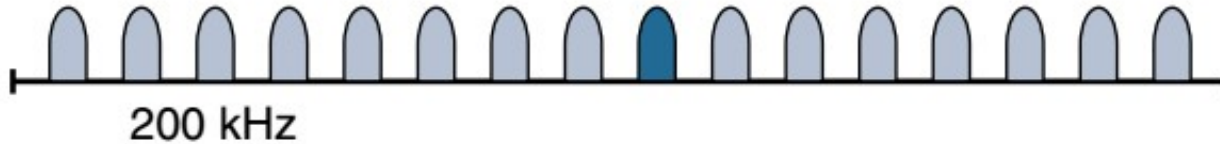
Three modes of operation are applicable to NB-IoT

- **Standalone:** A GSM carrier is used as an NB-IoT carrier, enabling reuse of 900 MHz or 1800 MHz.
- **In-band:** Part of an LTE carrier frequency band(1920 MHz - 2170 MHz) is allocated for use as an NB-IoT frequency. The service provider typically makes this allocation, and IoT devices are configured accordingly
- **Guard band:** An NB-IoT carrier is between the LTE or WCDMA (2100 MHz) bands. This requires coexistence between LTE and NB-IoT bands

NB-IoT - Deployment Options

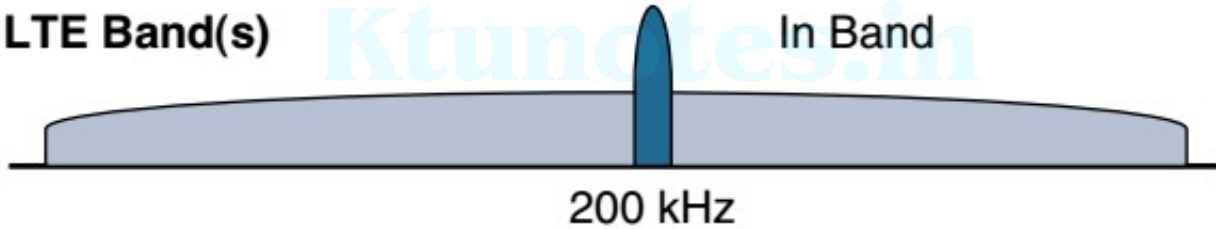
GSM Band(s)

Standalone



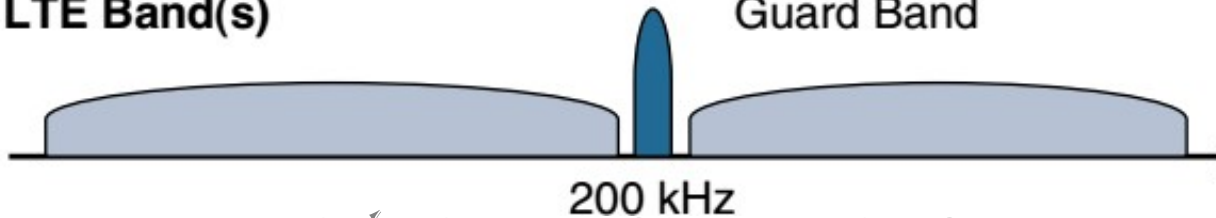
LTE Band(s)

In Band



LTE Band(s)

Guard Band



NB-IoT - Deployment Options

Mobile service providers consider NB-IoT the target technology as it allows them to leverage their licensed spectrum to support LPWA use cases.

For instance, NB-IoT is defined for a **200-kHz**-wide channel in both uplink and downlink, allowing mobile service providers to optimize their spectrum, with a number of deployment options for GSM, WCDMA, and LTE spectrum

NB-IoT

- NB-IoT operates in **half-duplex frequency-division duplexing** (FDD) mode with a maximum data rate uplink of 60 kbps and downlink of 30 kbps
- In an LTE network, resource blocks are defined with an effective bandwidth of 180 kHz, while on NB-IoT, tone or subcarriers replace the LTE resource blocks.
- The uplink channel can be 15 kHz or 3.75 kHz or multi-tone ($n \times 15$ kHz, n up to 12).

IP Based Protocols

Advantages of the IP suite for the Internet of Things

1. Open and standards-based

- Users can leverage a **large set of devices** and **functionalities** while guaranteeing **interchangeability** and **interoperability, security, and management** - open, standards-based solutions
- The IETF (Internet Engineering Task Force) is an open standards body that focuses on the development of the Internet Protocol suite and related Internet technologies and protocols.

IP Based Protocols

Advantages of the IP suite for the Internet of Things

2. Versatile

- A **large spectrum** of access technologies is available to offer connectivity of “things”
- The layered IP architecture is well equipped to cope with **any type** of physical and data link layers
- This makes IP ideal as a **long-term investment** because various protocols at these layers can be used in a deployment **now and over time, without** requiring **changes** to the whole solution **architecture** and **data flow**

IP Based Protocols

Advantages of the IP suite for the Internet of Things

3. Ubiquitous

- All recent operating system releases, from general-purpose computers and servers to lightweight embedded systems (TinyOS, Contiki, and so on), have an integrated dual (IPv4 and IPv6) IP stack that gets enhanced over time
- IP is the most pervasive protocol considering what is supported across the various IoT solutions and industry verticals

IP Based Protocols

Advantages of the IP suite for the Internet of Things

4. Manageable and highly secure

→ One of the benefits that comes from 30 years of operational IP networks is the well-understood network management and security protocols, mechanisms, and toolsets that are widely available.

5. Stable and resilient

6. Consumers' market adoption

IP Based Protocols

Adoption or Adaptation of the Internet Protocol

- **Two** possible **models** while IP based protocols are used in IoT
- **Adaptation** means application layer gateways (**ALGs**) must be implemented to ensure the **translation** between **non-IP** and **IP** layers.
- **Adoption** involves **replacing** all **non-IP layers** with their **IP layer counterparts**, simplifying the deployment model and operations
- Eg. ZigBee solution that runs a non-IP stack between devices and a ZigBee gateway that forwards traffic to an application server

IP Based Protocols

Factors to determine which model is best suited

→ **Bidirectional versus unidirectional data flow**

Unidirectional devices like a **switch**, it is **not necessarily** worth implementing a **full IP stack**

→ **Overhead for last-mile communications paths**

If **data overhead** in the form of extra header (20- TCP, 8- UDP) **cannot be tolerated**, IP **adoption** model is **not a good** option

IP Based Protocols

Factors to determine which model is best suited

→ Data flow model

IP **adoption** model works on the **end-to-end** nature of communications - Any node can easily exchange data with any other node.

In many IoT solutions, a device's data flow is limited to one or two applications (**not end to end**). In this case, the **adaptation model** can work

IP Based Protocols

Factors to determine which model is best suited

→ Network diversity

One of the **drawbacks** of the **adaptation** model is a general **dependency** on **single PHY and MAC layers**. For example, ZigBee devices must only be deployed in ZigBee network islands.

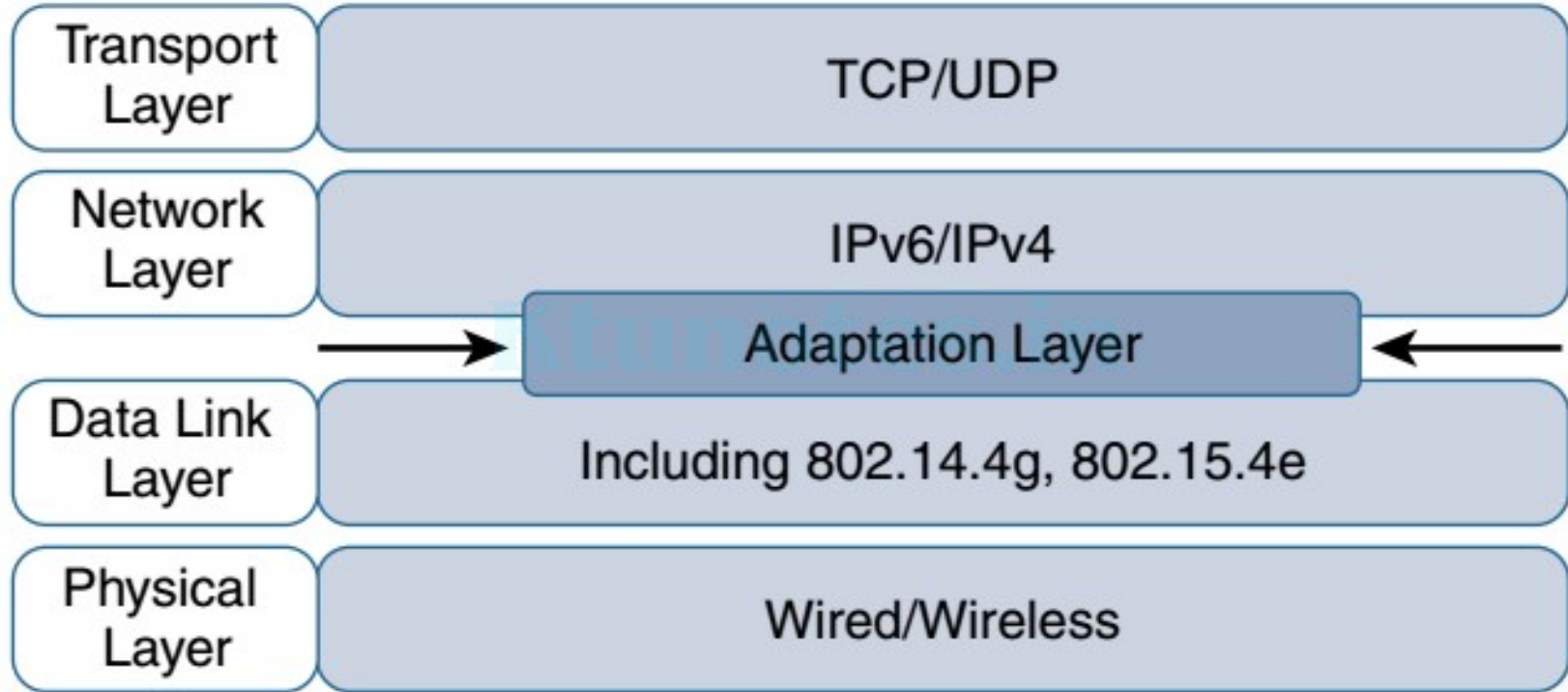
Therefore, a deployment must consider which applications have to run on the gateway connecting these islands and the rest of the world.

This is not a relevant consideration for the adoption model.

Optimizing IP for IoT

- While the Internet Protocol is key for a successful Internet of Things, constrained nodes and constrained networks mandate optimization at various layers and on multiple protocols of the IP architecture
- The **model for packaging IP into lower-layer protocols** is often referred to as an **adaptation layer**

Optimizing IP for IoT



Optimizing IP for IoT

- Unless the technology is proprietary, **IP** adaptation layers are typically **defined** by an **IETF** working group and released as a Request for Comments (RFC)
- An **adaptation layer** designed for IoT may include some **optimizations** to deal with **constrained** nodes and networks
- The main examples of adaptation layers optimized for constrained nodes or “things” are the ones under the **6LoWPAN** (IPv6 over Low-Power Wireless Personal Area Networks) working group and its successor, the **6Lo** working group

Optimizing IP for IoT

Comparison of an IoT Protocol Stack Utilizing 6LoWPAN and

an IP Protocol Stack - The 6LoWPAN protocol does not support IPv4

IP Protocol Stack

HTTP		RTP	
TCP	UDP	ICMP	
IP			
Ethernet MAC			
Ethernet PHY			

Application

Transport

Network

Data Link

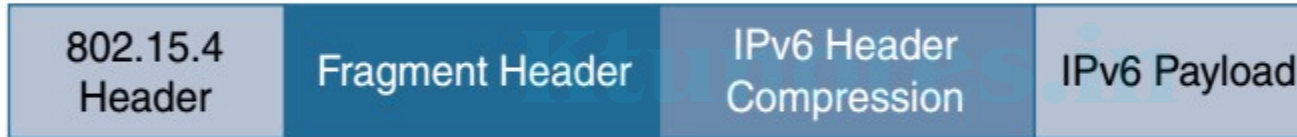
Physical

**IoT Protocol Stack with
6LoWPAN Adaptation Layer**

Application Protocols	
UDP	ICMP
IPv6	
LoWPAN	
IEEE 802.15.4 MAC	
IEEE 802.15.4 PHY	

Optimizing IP for IoT

6LoWPAN header stack variations



Optimizing IP for IoT

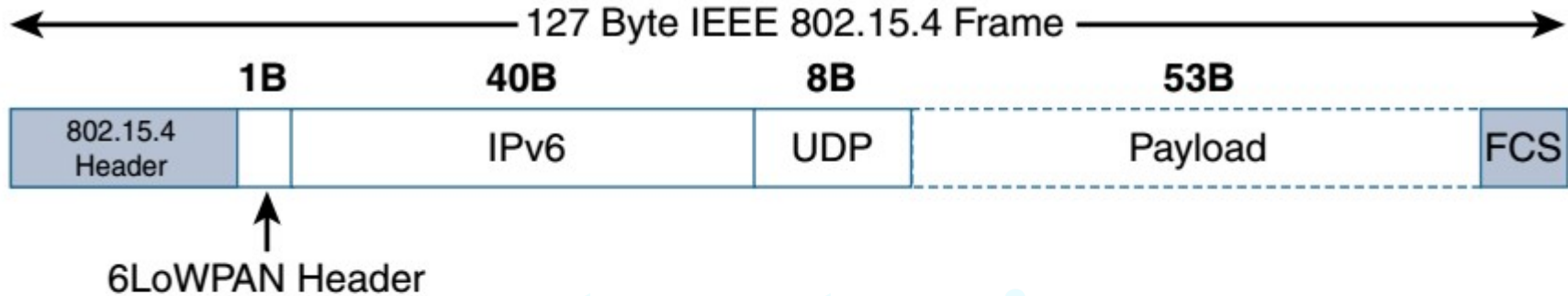
6LoWPAN header stack variations

→ Header Compression

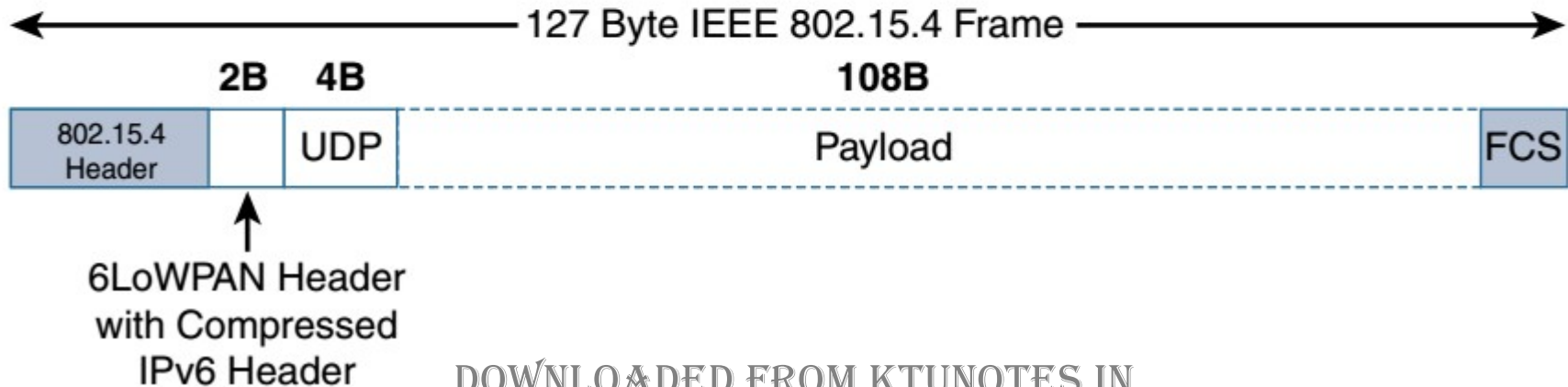
- This capability **shrinks** the size of **IPv6's 40-byte headers** and User Datagram Protocol's (UDP's) 8-byte headers down as low as **6 bytes** combined
- Reduction in header size is **implemented** by **omitting** some **standard header fields** by **assuming commonly used values** and by taking advantage of **shared information known by all nodes** from their participation in the local network

Optimizing IP for IoT

6LoWPAN Without Header Compression



6LoWPAN With IPv6 and UDP Header Compression



Optimizing IP for IoT

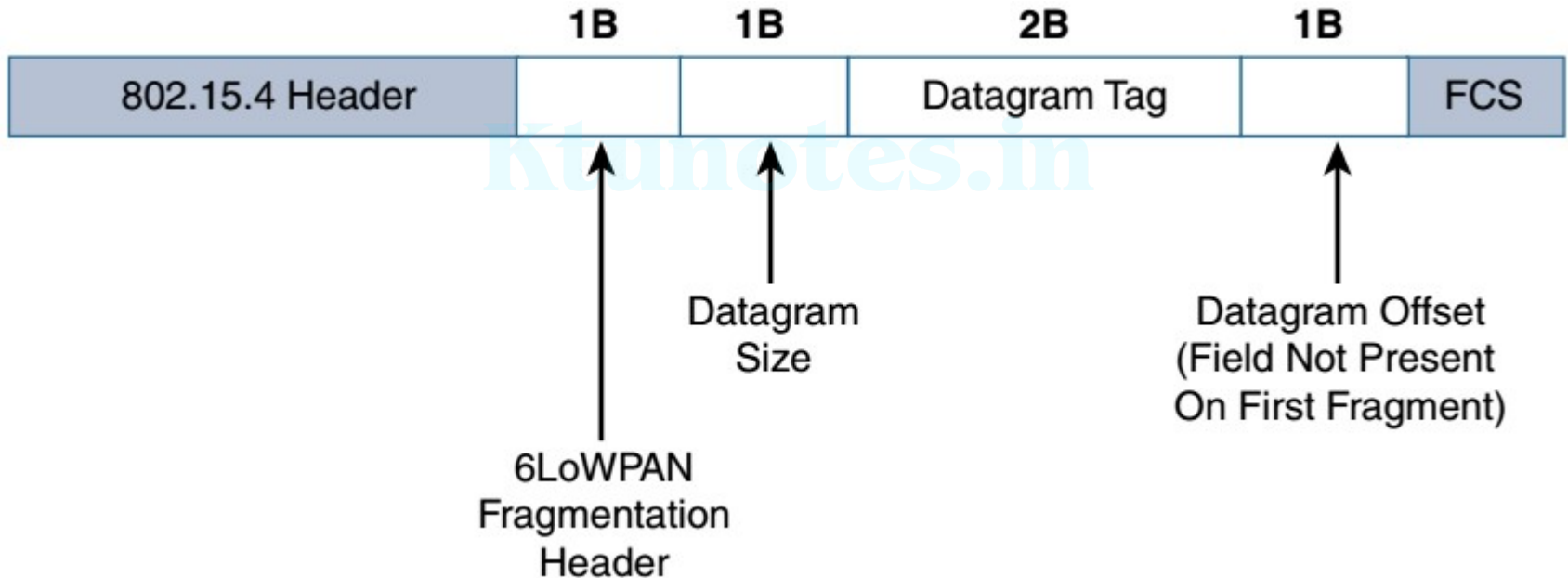
Fragmentation

- The **maximum transmission unit** (MTU) for an IPv6 network must be **at least** 1280 bytes
- For IEEE 802.15.4, 127 bytes is the MTU
- IPv6, with a much larger MTU, is carried inside the 802.15.4 frame with a much smaller one
- To remedy this situation, **large IPv6 packets** must be **fragmented** across multiple 802.15.4 frames at Layer 2
- The fragment header utilized by 6LoWPAN is composed of three primary fields -

Optimizing IP for IoT

Fragmentation

→ Datagram Size, Datagram Tag, and Datagram Offset



Optimizing IP for IoT

Fragmentation

- The 1-byte Datagram Size field specifies the total size of the unfragmented payload.
- Datagram Tag identifies the set of fragments for a payload.
- Datagram Offset field delineates how far into a payload a particular fragment occurs
- Fragmentation header field itself uses a unique bit value to identify that the subsequent fields behind it are fragment fields as opposed to another capability, such as header compression

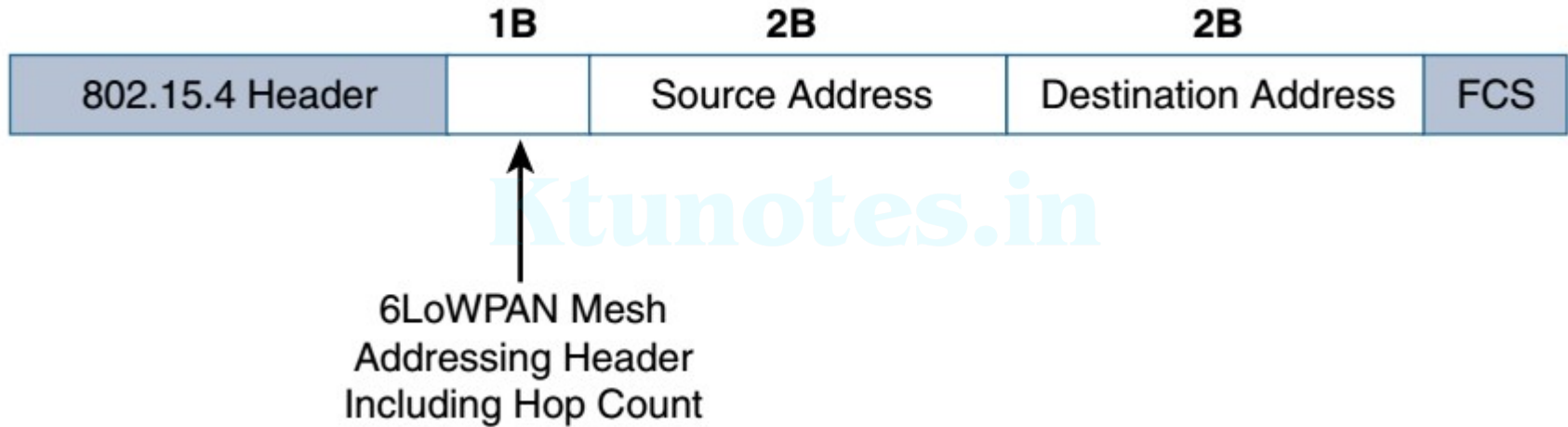
Optimizing IP for IoT

Mesh Addressing

- The purpose of the 6LoWPAN **mesh addressing** function is to **forward packets over multiple hops**.
- Three fields are defined for this header: **Hop Limit**- provides an upper limit on how many times the frame can be forwarded - Each hop decrements this value by 1 as it is forwarded. Once the value hits 0, it is dropped and no longer forwarded
- , **Source Address**, and **Destination Address**

Optimizing IP for IoT

Mesh Addressing



RPL - Routing Protocol for Low Power and Lossy Networks

- It is a new **distance-vector routing** protocol for **IPv6** for use by IP **smart objects**
- In an RPL network, **each node acts as a router** and becomes part of a mesh network.
- Routing is performed at the IP layer.
- **Each node examines every received IPv6 packet** and determines the next-hop destination based on the information contained in the IPv6 header.
- **No information from the MAC-layer header is needed** to perform **next-hop** determination

RPL - Routing Protocol for Low Power and Lossy Networks

- To cope with the **constraints** of **computing** and **memory** the protocol defines two modes
1. **Storing mode:** **All nodes contain the full routing table** of the RPL domain. Every node knows how to directly reach every other node
 2. **Non-storing mode:** **Only the border router(s)** of the RPL domain contain(s) the full routing table. All other nodes in the domain only maintain their list of parents and use this as a list of default routes toward the border router. This abbreviated routing table saves memory space and CPU

RPL - Routing Protocol for Low Power and Lossy Networks

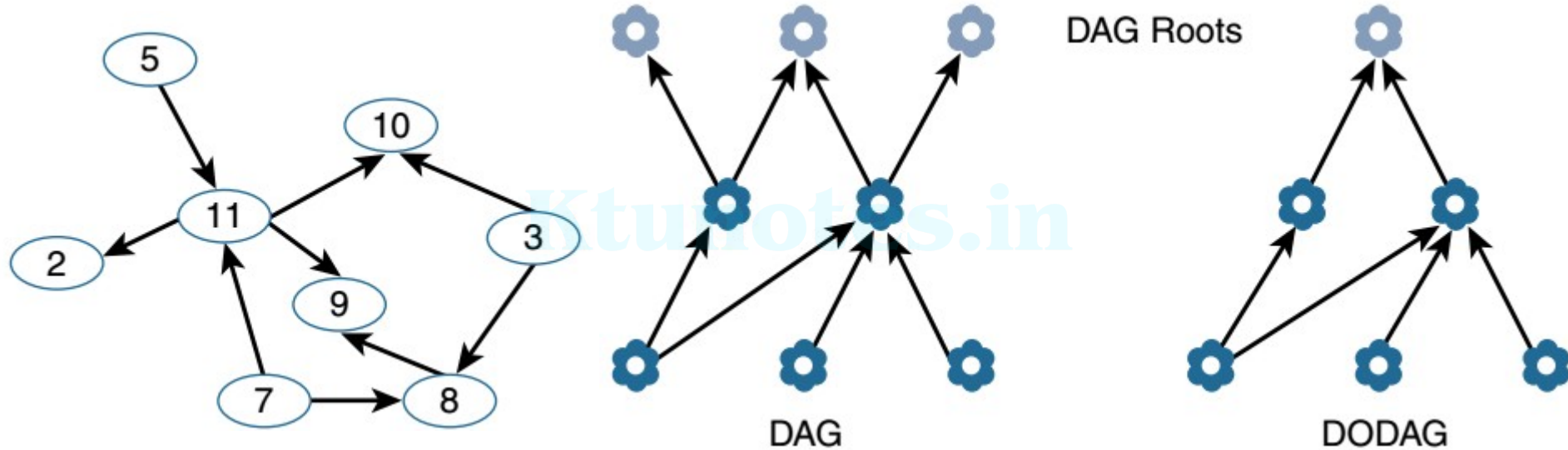
Non-storing mode:

Here a **node always forwards its packets to the border router**, which knows how to ultimately reach the final destination

- RPL is based on the concept of a directed acyclic graph (DAG). A DAG is a directed graph where no cycles exist.
- This means that from any vertex or point in the graph, you cannot follow an edge or a line back to this same point.
- All of the edges are arranged in paths oriented toward and terminating at one or more root nodes

RPL - Routing Protocol for Low Power and Lossy Networks

A Directed Acyclic Graph (DAG)



RPL - Routing Protocol for Low Power and Lossy Networks

Routing Process

- A basic RPL process involves building a destination-oriented directed acyclic graph (DODAG).
- A DODAG is a DAG rooted to one destination.
- DAG has multiple roots, whereas the DODAG has just one
- In RPL, this destination occurs at a border router known as the DODAG root
- In a DODAG, each node maintains up to three parents that provide a path to the root

RPL - Routing Protocol for Low Power and Lossy Networks

Routing Process

- Typically, one of these parents is the preferred parent, which means it is the preferred next hop for upward routes toward the root
- The routing graph created by the set of DODAG parents across all nodes defines the full set of upward routes.
- RPL protocol implementation should ensure that routes are loop free by disallowing nodes from selected DODAG parents that are positioned further away from the border router.

RPL - Routing Protocol for Low Power and Lossy Networks

Routing Process

- Upward routes in RPL are discovered and configured using DAG Information Object (DIO) messages.
- Nodes listen to DIOs to handle changes in the topology that can affect routing.
- The information in DIO messages determines parents and the best path to the DODAG root

RPL - Routing Protocol for Low Power and Lossy Networks

Routing Process

- Nodes establish downward routes by advertising their parent set toward the DODAG root using a Destination Advertisement Object (DAO) message.
- DAO messages allow nodes to inform their parents of their presence and reachability to descendants

RPL - Routing Protocol for Low Power and Lossy Networks

Routing Process

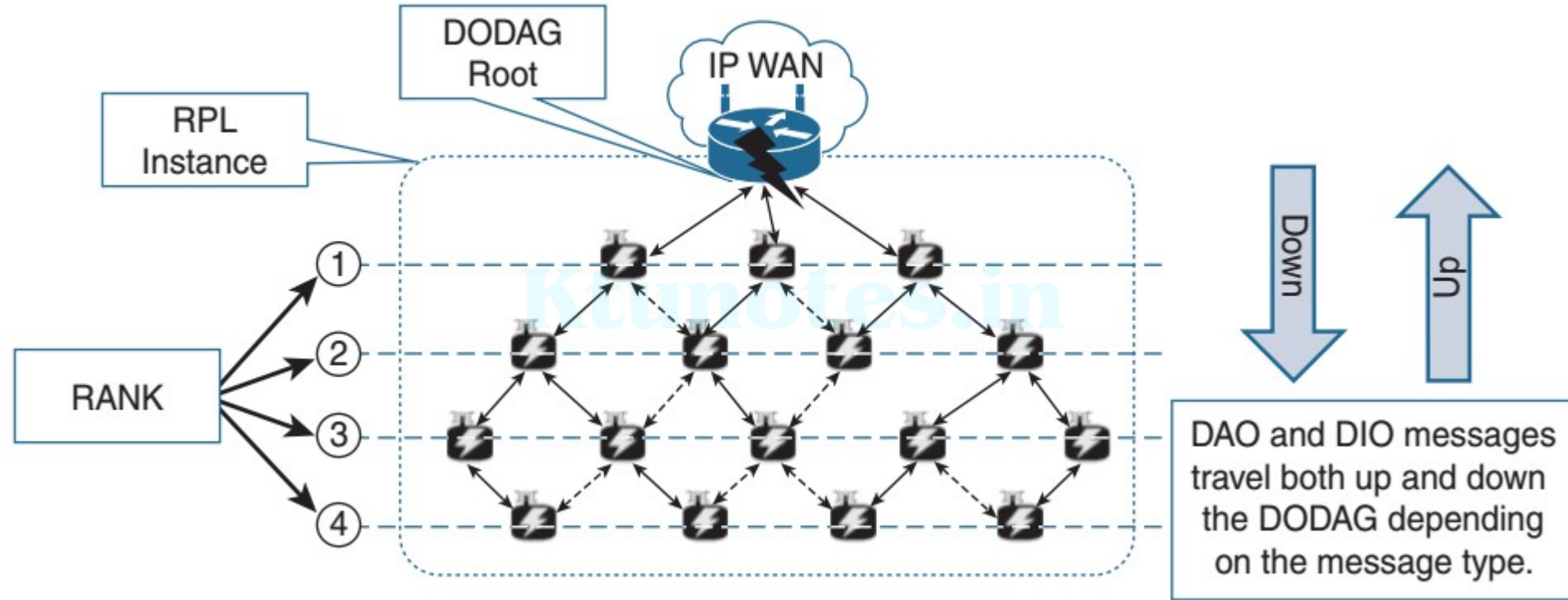
- In the case of the non-storing mode of RPL, nodes sending DAO messages report their parent sets directly to the DODAG root (border router), and only the root stores the routing information.
- The root uses the information to then determine source routes needed for delivering IPv6 datagrams to individual nodes downstream in the mesh

RPL - Routing Protocol for Low Power and Lossy Networks

Routing Process

- For storing mode, each node keeps track of the routing information that is advertised in the DAO messages.
- While this is more power- and CPU-intensive for each node, the benefit is that packets can take shorter paths between destinations in the mesh.
- The nodes can make their own routing decisions; in non-storing mode, on the other hand, all packets must go up to the root to get a route for moving downstream

RPL - Routing Protocol for Low Power and Lossy Networks



RPL - Routing Protocol for Low Power and Lossy Networks

- An **objective function** (OF) defines how metrics are used to select routes and establish a node's rank
- The **rank** is a rough approximation of how “close” a node is to the root and helps avoid routing loops and the count-to-infinity problem.
- Nodes can only increase their rank when receiving a DIO message with a larger version number.
- However, nodes may decrease their rank whenever they have established lower-cost routes

RPL - Routing Protocol for Low Power and Lossy Networks

RPL Metrics

- **Expected Transmission Count (ETX):** Assigns a discrete value to the number of transmissions a node expects to make to deliver a packet
- **Hop Count:** Tracks the number of nodes traversed in a path. Typically, a path with a lower hop count is chosen over a path with a higher hop count.
- **Latency:** Varies depending on power conservation. Paths with a lower latency are preferred

RPL - Routing Protocol for Low Power and Lossy Networks

RPL Metrics

- **Link Quality Level:** Measures the reliability of a link by taking into account packet error rates caused by factors such as signal attenuation and interference.
- **Link Color:** Allows manual influence of routing by administratively setting values to make a link more or less desirable
- **Node State and Attribute:** Identifies nodes that function as traffic aggregators and nodes that are being impacted by high workloads

RPL - Routing Protocol for Low Power and Lossy Networks

RPL Metrics

- Node Energy: Avoids nodes with low power, so a battery-powered node that is running out of energy can be avoided and the life of that node and the network can be prolonged.
- Throughput: Provides the amount of throughput for a node link. Often, nodes conserving power use lower throughput. This metric allows the prioritization of paths with higher throughput